

Pandora's Box Reopened: Can Generative AI Restore Hope or Result in a Decline in the Quest for Academic Integrity

Journal of Macromarketing
2025, Vol. 45(4) 831–862
© The Author(s) 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/02761467251363569
journals.sagepub.com/home/jmk



Jared M Hansen¹ , OC Ferrell², Linda Ferrell² , Michael A Levin³ , Paul Wilson⁴
and Kellie Cloud⁵

Abstract

This research outlines the potential impact of generative artificial intelligence (GenAI) on research integrity in marketing scholarship and other fields that macromarketing research draws on. The research provides a detailed analysis of the likely outcomes of GenAI on already declining research integrity, the probable outcomes in the future if the trends continue, and a roadmap for a better future in which GenAI enhances rather than undermines research integrity. Applying the H-V model of ethical decision-making, the research contrasts a range of viewpoints related to different belief statements of right and wrong (deontological norms) that people might draw upon in determining which activities involving GenAI at different stages of a research paper development are proper to pursue. The authors emphasize macromarketing's responsibility to establish guard rails for responsible GenAI use in scientific research to maintain trust, advance knowledge, and seek truth.

Keywords

generative artificial intelligence (GenAI), research integrity, predatory journals, retractions, Hunt-Vitell (H-V) model of ethical decision making, hope, truth

Although our world is verging on dramatic societal and economic shifts, business in general – and marketing in particular – seems unprepared for them.

– Hyman and Kostyk (2019, p. 1485)

A lie can travel around the world and back again while the truth is lacing up its boots

–Mark Twain (1921)¹

Introduction

Academia currently shows significant fascination with generative artificial intelligence (hereafter GenAI). GenAI is a relatively new subset of AI technology that, definitionally, produces ‘creative’ new content (e.g., text, audio, image, and video) based on patterns from training data. All GenAI is a form of AI, but not all AI is generative. AI in various forms and, by extension, the large language models (LLMs) that comprise them, have been around and affected marketing practice and consumers for years (see discussion in, e.g., Ameen et al., 2022; Kopalle et al., 2022; Ma & Sun, 2020; Mariani et al., 2022; Mehta et al., 2022). However, GenAI technology took center stage in both industry and society with the introduction of ChatGPT in November 2022. As stated by Christensen

et al. (2024, p. 545), “ChatGPT, the currently most popular generative AI software, was launched one year ago. Today, it is the fastest growing and one of the top visited websites in the world.” Several other popular GenAI tools have emerged since then.

In marketing alone, conferences sponsored by the American Marketing Association, Association for Consumer Research, Academy of Marketing Science, INFORMS, Society for Marketing Advances, Society for Consumer Psychology, Macromarketing Society, and European Marketing Academy promoted GenAI as a part of their respective recent conference themes during the writing of this article. Several marketing journals have posted calls for papers (CFPs) related to GenAI and marketing. Regarding GenAI, Davenport et al. (2020, p. 25) declare, “The marketing discipline should take a lead

¹Utah State University, USA

²Auburn University, USA

³University of Louisiana at Lafayette, USA

⁴Brigham Young University, Hawaii, USA

⁵University of North Carolina at Charlotte, USA

Corresponding Author:

Jared M Hansen, Associate Professor of Marketing, 3535 Old Main Hill, Logan, UT, 84322.

Email: jared@usu.edu

role in addressing these questions because arguably it has the most to gain.” A McKinsey industry report looking at “more than 400 use cases across 19 industries and nine business functions” predicts AI’s impact to be most substantial in marketing and supply-chain management areas (Chui et al., 2018, para. 1). Ooi et al. (2025) also outline several areas in which GenAI can impact business; many of the areas fall under the umbrella of marketing functions. Hyman and Lukosius (2024, p. 915) state, “AI in marketing will transform society and marketing practice.” See also, e.g., Grewal et al. (2024b) and Mu and Zhang (2025). We agree with these statements that marketing remains an essential place for discussing GenAI.

Furthermore, we note that GenAI could affect macromarketing research differently than it affects other disciplines in a couple of important ways. First, marketing, in general, and macromarketing, in particular, have an interdisciplinary draw on medicine and health science, psychology, economics, hard sciences, and other disciplines and domains (Ekici et al., 2021)—which can result in a cascading effect of GenAI based integrity issues from the improper use of AI in publications in those fields on marketing research. Second, marketing research often involves surveys and consumer experiments that face challenges from synthetic data from GenAI that researchers could have difficulty detecting. Third, GenAI is potentially changing what and how consumers think, as they use the tool in searching for information and making decisions instead of using traditional search engines. Fourth, marketing as a discipline spans many questions that often contain ethical dimensions or content (Ferrell, 2021; Ferrell & Ferrell, 2022; Gundlach & Murphy, 1993; Hunt & Vitell, 1986; Lacznik & Murphy, 2006, 2012; Murphy, 2017; Schlegelmilch & Öberseder, 2010; Shabbir et al., 2021). We strongly caution that *GenAI is incapable of making moral decisions because it cannot feel regret or remorse. Thus, using it as an aid in addressing all of the marketing questions that contain ethical content needs to be done with great care and awareness.* Given all of the preceding, marketing science has an imperative responsibility to not just ‘sit at the table’ (Day & Malcolm, 2012; Moorman & Day, 2016; Reibstein et al., 2009; Venkatesh & Peñaloza, 2015), but to develop models and establish guard rails in discussing responsible AI, including GenAI, for all areas of scientific research.

A growing stream of studies has focused on how professors and universities could combat *students’* unethical usage of GenAI in course assignments (e.g., Crawford et al., 2023; Ibrahim et al., 2023; Kenwright, 2024; Rane et al., 2023). Unfortunately, all efforts to combat unethical usage seem to be failing. A separate set of studies has focused on different concerns regarding GenAI, which can include bias in either the algorithms or the training data, accuracy vs. errors in interpretation of data, privacy and attribution/copyright, or debates over using the terms: “hallucination,” “non sequitur,” or “unrelated response” (Østergaard & Nielbo, 2023) or “bullsh!t” (Hicks et al., 2024); see, e.g., the review in Sharma (2024, pp. 5–6) of several recent studies that focus on these types of

quality and truth concerns. In this paper, however, we take a different focus from those important, growing areas of research. We focus on GenAI’s impact on academic research by scholars and industry professionals.

The purpose of this research is to outline (a) how GenAI can *amplify* the spiraling integrity crisis in scholarly research by researchers (not students), (b) what are some solutions to avoid the further acceleration of (or perhaps even diminish or nearly eliminate) the problem, and (c) how GenAI could instead be used to improve the integrity and insights of research. We believe that the impact of GenAI on academic marketing research integrity will not be constrained to individual ethical lapses of researchers; rather, the impact is potentially a systemic issue that will permeate the discipline. Strong evidence suggests that the pressure to publish and the widespread availability of GenAI tools create the opportunity for misconduct in scientific publishing. In response, this research article describes how elements of ethical decision-making frameworks appearing in the *Journal of Macromarketing*, such as the Hunt-Vitell (1986, 2006) model, can be combined in conjunction with concepts from Jones’ (1991) moral intensity model to provide a theoretical framework for guiding GenAI usage. We combine the results of an expansive literature review and personal observation, outlining both possible negative and desirable positive futures given the probable increases in GenAI usage. We predict that there will be technological advances made in GenAI by the time readers see this article that did not exist when we wrote it. However, we posit that the core premises, warnings, opportunities, and guidelines will continue to be relevant for the next several years.

The rest of this research proceeds as follows. First, we outline current challenges to academic research integrity, including publication pressure challenges, submission increases challenges, data fabrication/misconduct challenges, and retraction challenges. We discuss how GenAI might amplify these situations. Second, we predict probable outcomes if the status quo is maintained. Third, we describe the impact on marketing research, policy, and consumers. Fourth, we describe potential future integrity-related challenges, with a focus on the fraud triangle, synthetic datasets, and smoke and mirrors syndrome. Fifth, we discuss the roles of journals’ publishing codes of ethics. Sixth, we outline several potential challenges in the near future related to GenAI and scholarship: data integrity, cognitive warfare, social isolation of researchers, equity in grants, narrowing of individual researchers’ knowledge growth, and complex moral issues. Seventh, we outline several research questions about the morality of using GenAI in several steps in the research process through the lens of the Hunt Vitell (H-V) model of ethical decision-making. Last, we list several possibilities for the future and then provide conclusions.

Challenges to Academic Research Integrity

Reviewing current issues related to research integrity in publishing is essential. While the focus of this paper centers on GenAI as

a tool to support academic research, framing the types of moral misconduct linked to the inappropriate use of GenAI is also important to discuss, as outlined by Jones (1991), to establish the intensity, concreteness, and proximity of the moral intensity of the issue. This section addresses intentional fraud in scientific research. We first outline the current, growing concern about academic research integrity. We describe challenges related to (a) publication pressures, (b) submission increases at journals, (c) data fabrication and misconduct, and (d) retractions.

Publication Pressure Challenges

Ross (2015, para. 25) states that “Rising expectations, coupled with diminished resources to support professional activity, have increased pressure on pre-tenure faculty.” This pressure to publish, which exists at nearly every higher education institution across the globe, has resulted in a more aggressive application of tools to facilitate success ‘at any cost’ in publishing by some scholars. Some universities are facilitating and incentivizing publication with excessive cash payouts for hitting certain level journals. For example, we were recently informed that researchers at some Chinese universities now receive awards of up to \$75,000 for publishing in A-ranked journals.

At many institutions, publishing pressure is so intense that some professors might believe no alternatives exist for them to keep their university affiliation and status other than engaging in misconduct. A recent survey of over 1,300 faculty at over 500 universities revealed that “More administrative work is being passed down to faculty” (Cengage, 2024, p. 10) and that “among our dissatisfied faculty respondents, not having enough time to conduct research was among the top five dissatisfaction drivers for the first time in our survey’s history” (Cengage, 2024, p.11). Coupled with burnout from the recent pandemic, these workload increases can result in greater pressures to compromise research integrity while striving to retain university employment.

Additionally, other professors are engaging in misconduct to facilitate acquiring citizenship; recent controversy has emerged over the blog by Debarghya Das, a tech influencer and former Google engineer who recently encouraged researchers to “commit scientific fraud” and detailed which ‘pay to play’ journals are the more versus less expensive routes and take the least time from submission to publication (often less than a month from initial submission to publication) in his ‘ultimate guide-book’ to get United States citizenship (e.g., Gonsalves, 2024; Times of India, 2024).

In review of the preceding factors, we argue that the culture for publishing has evolved without appropriate education and training to systematically encourage good academic citizenship behaviors that prevent misconduct. The emphasis has been on the publishing outcomes at the potential cost of integrity in the process. Not enough people in the marketing academy are taking a leadership role in helping to mitigate this current situation. At the same time, this issue extends beyond marketing; the entire academic ecosystem faces similar challenges and unfortunate outcomes regarding research integrity.

Submission Increases and Quality Challenges

Complicating the job of editors and reviewers examining submissions for research integrity is the increasing number of submissions to many journals. For example, Donthu et al. (2021) report that the number of submissions to the *Journal of Business Research* (JBR) increased from approximately 2,000 submissions in 2013 to 5,000 submissions in 2019 and 7,000 submissions in 2020. See also Grewal (2017). Those reported increases were pre-pandemic. The COVID-19 pandemic led to an even greater surge in submissions to many journals. For example, JBR reported a 105% increase in submissions from January to April 2020 compared to the same period in 2019 (Donthu et al., 2021). Post-pandemic submissions have continued to increase at an accelerating rate. As a result, many editors and reviewers are almost overwhelmed and journals have to increase the number of co-editors and associate editors, review boards, and ad hoc reviewers to work through the increased workload.

Much of the more recent increases in submissions to North American and European-based scientific research journals appear to be coming from scholars located predominantly in Asia, Russia and India.² According to a United States National Science Foundation report about science and engineering publications:

“The growth in worldwide annual publication output (from 2.0 million in 2010 to 3.3 million in 2022) was driven in particular by two countries: China (42% of additional publications during that period) and India (11%) together accounted for more than half of that increase in publications...Russia, South Korea, Iran, and Brazil made notable contributions to the growth in the number of publications from the rest of the world from 2010 to 2022,” (Schneider et al., 2023, para. 3).

This reported shift in submission origin is significant given the finding by Sabel et al. (2023) that the growing data integrity and retraction problem is not globally uniform. Sabel et al. report in the analysis about retractions for plagiarism (RFPs) that the “Countries with the highest RFP proportion were Russia, Turkey, China, Egypt, and India (39%-55%)” (Sabel et al., 2023, p. 1). Those differences in RFPs across geographies appear consistent with the patterns found in other research over the last two decades on academic student dishonesty. For example, Maloshonok and Shmeleva (2019) and Magnus et al. (2002) both find that students in Russia are more tolerant of cheating than students in the United States. Grimes (2004), Rawwas et al. (2004), Fass-Holmes (2018), Payan et al. (2010), and Bratton and Strittmatter (2013) find similar attitudes (of more acceptance of cheating) held by students in China. See also reviews in, e.g., Hu and Lei (2015), Hu and Lei (2016), Liu and Chen (2018), Jian et al. (2019), and Pan and Lei (2024). Additionally, Rawwas et al. (2006) and Crittenden et al. (2009) find that students in India are also more tolerant of cheating than students in America.³ We believe that these different geographic norms regarding academic honesty and originality of research extend beyond students.

Cabanac et al. (2023, p. 1) review 12,437 retracted papers from the Crossref DOI (digital object identifier) registration agency. They find that “analysis of the geographic distribution of retracted papers reveals a concentration in Asia that is increasing over time.” Yang (2013, p. 1019) states, “China’s research capacity has grown dramatically in the past decade...Most troubling is a lack of research integrity, which may hinder China’s growth in original science, damage the reputation of Chinese academics, and dampen the impact of science developed in China.” A posted response to the Yang article online by Guan (2016) states, “It is estimated that over 90% [of] papers in China are ‘rubbish.’ These misconducts retard the advancement of academic research in China.” Zhang (2023) reports that it is not uncommon for doctoral students in China to face threats from their tutors (aka dissertation chair/advisor) regarding funding or graduation when they speak out. Tian and Lu (2017, p. 957) report that there is “intensified academic pressure and consequent feelings of insecurity, uncertainty, and anxiety.” Given that situation, it should not be surprising that Liu and Chen (2018) report that “Compared with publicized cases of falsified or fabricated data by authors from other countries of the world, the number of Chinese academics exposed for research misconduct has increased dramatically in recent years.”

Regarding business and management journals, in particular, Xu et al. (2018) report that submissions from Asia to business and management journals increased close to 200% between 2004 and 2013, while submissions from North America increased only about 30% during the same period. More Asian universities now require their faculty to publish in elite journals based in the United States and Europe. For example, a decade ago, the Chinese government established specific targets for increasing the number of research articles published in top international journals by Chinese researchers (Cao et al., 2013; Cyranoski, 2018; Xie et al., 2014). In support of those goals, many Chinese universities have implemented reward systems, including very significant financial bonuses for researchers who publish in high-impact journals (Quan et al., 2017). As a result, China’s percentage of scientific publications in elite journals has grown significantly, surpassing the United States in the total number of science and engineering publications a few years ago (Schneider et al., 2023).⁴

Data Fabrication and Misconduct Challenges

Trust in the truth of scientific research is facing a growing crisis due to problems with data fabrication and other misconduct (Oransky & Redman, 2024). For instance, trade press news articles now more often feature high-profile researchers, such as Diederik Stapel, Dan Ariely, Francesca Gino, Brian Wansink, Lawrence Sanna, Hans-Ulrich Wittchen, Khalid Shah, and Dirk Smeesters, among others, who are accused of fabricating data or results (e.g., Borsboom & Wagenmakers, 2012; Boytchev, 2021; Callaway, 2011; Crocker & Cooper, 2011; Dahlberg, 2018; Jha, 2012; Lewis-Kraus, 2023; O’Connor, 2018; Ravi & Paulus, 2024; Stern, 2023; Yong, 2012) and

leaders of colleges and universities, such as Claudine Gay at Harvard, Marc Tessier-Lavigne at Stanford, and Erick Jones at the University of Nevada at Reno who resigned in response to accusations of plagiarism or data fabrication (Baker, 2023; Gelman, 2024; Moody, 2024). While it could be argued that some instances of research integrity investigations have been weaponized (Bailey, 2024), these high-profile investigations erode the public’s trust in higher education and the corresponding members of the academy. Indeed, a few years ago, a Gallup poll found that there has been a tremendous drop in public confidence in higher education over the past several years in the United States, which is “more so than for any other U.S. institution that Gallup measures” (Marken, 2019).⁵ The most recent Gallup poll indicates that trust in higher education has continued to decline at an increasing rate (Nietzel, 2024).

On the other hand, there appear to be numerous instances of research articles with fabricated data that have remained unknown to most marketing scholars and are not highlighted, yet collectively impact the integrity of science and research as a whole. The issue appears acute in medical research. For example, Piper (2024) states that “for years, biomedical researchers have been aware that the field has a problem with faked images in papers.” Bik et al. (2016) also discuss a similar problem. Thiese et al. (2017, p. 4117) note that “reported prevalence estimates of misconduct are probably underestimates” in the medical field. Buntz (2023) outlines nearly three dozen unique major medical advances over the last few decades that were fabricated. In their analysis, Sabel et al. (2023) conclude in a medRxiv preprint that a significant proportion of research studies are falsified or plagiarized in biomedical journals (24% of studies) and neuroscience journals (34% of studies) looking just at 2020 studies. David Stewart, former editor of multiple elite marketing journals, indicated while speaking on a panel at the 2024 AMA Public Policy and Marketing conference that research in about 25% of studies in medical journals is fabricated.⁶

Retraction Challenges

The amount of academic research data integrity issues is now staggering. And with those issues comes the next challenge: retractions. Many journals are retracting research articles due to misconduct (Cabanac et al., 2023). At one time, retraction counts were small. However, Van Noorden (2023, p. 479) states, “More than 10,000 research papers were retracted in 2023 — a new record.”

Fang et al. (2012) calculate that about two-thirds of the retractions are due to fraud or other misconduct. Websites such as retractionwatch.com list dozens to hundreds of ongoing investigations at any given time. Unfortunately, retractions often take years from the time of investigation before they officially occur. Sometimes, it can be a decade or longer before investigations into these articles begin. Concepts in retracted papers can continue to be spread through word of mouth long after the retraction occurs, such as the examples outlined in the next section related to vaccines and to associations between food consumption and health.

The Impact on Academic Marketing Research

While the previous statements are vividly disturbing, some might ask the question ‘But how do those problems affect us in marketing?’⁷ Or, put another way, why should we spend time in academic *marketing* research discussing those widespread problems in science? Jones (1991) indicates that ‘proximity’ (in addition to vividness) is an essential element of individuals’ ability to recognize that a decision contains moral issues. In this section we provide a few examples of specific exploration of the global problem identified in the previous section on macromarketing research topics.

Ekici et al. (2021) conducted a content analysis of all articles in the *Journal of Macromarketing* over the last decade. In the list of societal development categories of macromarketing research in their Table 3, an important category of macromarketing research is on health-related “wicked problems such as obesity, smoking, alcohol consumption” (Ekici et al., 2021, p. 29). These categories of marketing research topics draw heavily on research from medical fields. Many other marketing and consumer-based research studies also draw on research from medical studies or social psychology research. According to reports in the American Psychological Association: “Psychology has experienced a recent string of revelations of data fabrication and other lapses of scientific ethics” (Breckler, 2012). Many scholars accused of data or results fabrication in the first paragraph of this section of the paper are social psychologists who have had dozens of articles retracted from leading journals in psychology. Thus, the problem affects many categories of research in the marketing discipline.

For example, the article titled “Super Size Me” was published in 2012 in the *Journal of Consumer Research*. The authors requested a retraction 10 years later, in 2022, by which time there had been 368 articles citing the paper according to Google Scholar.⁸ Between 2023 and July 2024, there have been 59 new citations of the retracted article.⁹ Additionally, many other articles cited those 368 cited works or built on ideas from the retracted work without attribution. Retracted articles are still available but stamped as ‘retracted’ on publisher websites and on platforms like ResearchGate.

One reason for the continued citation after retraction is that some scholars engage in the common literature search practice of finding a few relevant research articles or dissertations and then citing papers from the reference section without reading the cited work. This means that they have no idea that the original citations might have possibly been retracted, and the original works *and/or ideas from them* continue to be cited. This practice seems consistent with the similar situation of consumers’ continual reposting of fake news on social media that has been fact-checked and proven false. These described concepts hurt data integrity.

The Impact on Policy and Legal Decisions Investigated in Macromarketing Research

The continued usage of ideas from retracted works is not just by scholars. The usage post retraction is also occurring among legal

decision-makers, an important stakeholder group of macromarketing research. For example, Sage retracted the articles Studnicki et al. (2019), Studnicki et al. (2021), and Studnicki et al. (2022) for conflict of interest, a problem with prepublication peer review, and unreliable findings. However, these three papers remain available through the journal’s website. Lawyers for the Alliance for Hippocratic Medicine cited these three papers in their court filing against the United States Food and Drug Administration in 2022, and the court trial judge referenced the three retracted studies in his ruling despite the papers not meeting the five-part test used by courts to determine whether to accept expert testimony outlined in the 1993 *Daubert v. Merrell Dow Pharmaceuticals* court case (i.e., 509 U.S. 579): “(1) Whether the technique or theory in question can be and has been tested; (2) Whether it has been subjected to publication and peer review; (3) Its known or potential error rate; (4) The existence and maintenance of standards controlling its operation, and (5) whether it has attracted widespread acceptance within a relevant scientific community” (Legal Information Institute, 2025).

No one knows how policymakers and legal decision-makers will remain current in the future with which ideas originated in research that has been retracted—recalling the record number of 10,000 retractions in 2023 mentioned earlier. The abilities of GenAI which “may fabricate nonexistent legal provisions, cases, or incorrect medical advice to answer questions, hindering judicial fairness and causing medical risks” for professionals in legal, medical, and other professional fields (Wang et al., 2024, p. 422), will only compound the problem of retracted or debunked research remaining in circulation and being treated as truth. Much of macromarketing research has public policy and legal implications and draws on research from public policy and legal reports.

The Impact on Consumers’ Decisions Investigated in Macromarketing Research

In this section, we outline a few examples of how integrity problems in research from other fields can impact consumers’ decisions focused on in macromarketing research. One topic of focus in macromarketing related research is vaccines (e.g., Akaka et al., 2023; Brush et al., 2024; Domegan et al., 2020; Park et al., 2025; Paygah & Colabi, 2023; Shabbir et al., 2021; Seo et al., 2023; Thyroff et al., 2023; Wang et al., 2023; Weber et al., 2021; Zimmermann et al., 2024). As to consumers drawing on retracted works containing research data integrity problems in their decision-making, one example related to vaccines is the research ideas presented by Wakefield et al. (1998). Twelve years after publishing Wakefield et al. (1998), the journal *Lancet* retracted the study. The paper claimed a link exists between loss of language development skills, gastrointestinal disease, and developmental regression to the measles, mumps, and rubella (MMR) vaccine—recalling that macromarketing is interested in health ‘wicked problems.’ Mass media repeated the paper’s claims over the twelve years, giving credence to the anti-vaccine movement in the United States and England (Motta & Stecula, 2021).

Journals can retract articles for issues related to conflict of interest, analysis, and data integrity. Once the ideas put forth in the research become available, however, they cannot simply be pulled back with a retraction because retracted articles remain available, even on the publisher's websites. These retracted articles are often at odds with societal good. For example, a victory for the plaintiffs would have eliminated the idea that federally approved medications would be available in the country. Instead, it became a question for states to approve. Anecdotally, Wakefield et al. (1998) are credited for the repeated rise in children in England and the United States who are not vaccinated (e.g., Breakwell, 2020, p. 512; Deer, 2020; Finkel & Rosen, 2007, p. 387; Stamatiadis-Bréhier, 2024, p. 12–13).

Other macromarketing-related studies focus on food safety regulations (e.g., Bech-Larsen & Aschemann-Witzel, 2012; Charlebois & Labrecque, 2009; Johnson-Hall, 2017; Latino et al., 2022; Manning, 2017; Newman et al., 2014; Truong, 2017). Within this research area, an example is the research associating monosodium glutamate (MSG) with migraines. As outlined in LeMesurier (2017) and Blanding (2019), the deception began in 1968 with a \$10 bet that Howard Steel, an orthopedic surgeon, and professor, made with another doctor, Bill Hanson, about whether he was capable of publishing in a prestigious journal such as the *New England Journal of Medicine* (NEJM). After writing the fake letter under a fake name, he was surprised when it was published in NEJM.

He reached out to the journal editor, letting him know of the satirical forgery, but was ignored. As reported by Blanding (2019), "As LeMesurier researched the journal, she discovered that NEJM actually had a long tradition of such "comic syndrome letters," as one observer called them: missives that used pretentious medical language to poke fun at a common problem." Blanding (2019) later states, "No matter how many articles appear exposing the fallacy of Chinese-restaurant syndrome, prejudice against MSG persists." What began as a \$10 joke bet changed the whole face of consumer and industry cooking and food processing. It has altered how consumers and medical professionals talk about a specific medical issue, migraines, and likely contributes to the belief that certain foods can profoundly alter our physical health.

The Negative Impact of GenAI on the Future of Research Integrity if We Maintain the Status Quo

While GenAI might assist scientists in accelerating the advancement of science, overreliance on GenAI could diminish originality and critical thinking, which are essential in academic research. Overreliance could also lead to a lack of accountability and the failure to develop new knowledge. Programs are being developed to evaluate sections where GenAI is used, rather than solely relying on the author's statements (Kardel et al., 2024). We have seen similar programs used to detect GenAI usage in student writing incorrectly flag original work

as AI-generated (e.g., Dalalah & Dalalah, 2023; Elkhatat et al., 2023). As the next generations of scholars spend their earlier educational years regularly using the GenAI tools, they could naturally write using the same style and tone that GenAI writes.

Thus far, this discussion has focused on what has transpired regarding research integrity over the last decade. Attention now shifts to emerging issues that, if left unaddressed, could become more significant problems. The preceding problems are accelerating, and the application of the capabilities of generative AI by authors who are unaware, untrained, uncaring, or intentionally malicious will make it much more difficult for legitimate scientific research to be effective at helping people, institutions, and society thrive and flourish. It is important to note that GenAI can be a valuable tool to help researchers in advancing science.

The authors of this research article use GenAI regularly for many activities related to research and other job functions. Many journal editors and publishers have recently stated at conferences that they like it when authors, especially authors who are writing in a second language, use GenAI to help them improve the accessibility and comprehension of their ideas. However, they do not want authors to use GenAI to write the manuscripts. The usage of GenAI to write manuscripts that appear credible is frightening. In the next subsections we review some examples.

The Integrity Challenges from the Existing Publishing Culture

It has been stated that one of the reasons that some researchers knowingly engage in unethical practices related to research integrity, at times adopting a Machiavellian teleological 'the ends justify the means' belief, is in response to the job pressure to publish or perish related to obtaining tenure or financial rewards (Herndon, 2016; Hunt & Chonko, 1984; Hyman, 2024; McQuarrie, 2014; van Dalen, 2021). These pressures now continue post-tenure with more and more institutions of higher education requiring faculty, instructors, clinical professors, and even professors of practice to maintain an academic qualification designation based largely on maintaining research-specific productivity. A loss of academic qualification can lead to a faculty member, regardless of rank, being stripped of tenure, demoted, and/or placed on probation culminating in employment termination.

The culture of publish or perish also exists in an environment where most academics have little or no training in publishing ethics. Doctoral programs provide scant guidance on publishing ethics, focusing instead on getting doctoral students published to get them on the job market. The publishing culture starts in doctoral programs, where students, with little guidance from their professors, learn from others how to get published.

The Fraud Triangle (Cressey, 1953) could explain intentional unethical publishing decisions (see, e.g., Latan et al., 2019; Rodgers et al., 2015). The Fraud Triangle includes three parts: (1) opportunity, (2) incentive, and (3) rationalization.

These three components can explain why people actively deceive. In an academic setting, opportunity would arise through the various GenAI tools and their corresponding applications, as discussed in this paper. The incentive rests in the forms of (a) getting a tenure track position or a professor of practice at a prestigious higher education institution, (b) publishing in a highly esteemed or well-regarded academic journal, or (c) receiving a promotion and/or publishing-based bonus. Finally, rationalization can be asserted that everybody else is doing it or I need to do it to get, keep, or be promoted at a job.

GenAI and the “Smoke and Mirrors” Syndrome

GenAI will continue to improve its abilities. Already, studies have shown that faculty struggle to distinguish papers written by GenAI from papers written by other faculty (Casal & Kessler, 2023). Harari (2016) argues that the pressure to produce groundbreaking research in combination with advanced technologies (such as GenAI) could lead to more sophisticated forms of academic fraud. One such form that is commonly referenced is having GenAI write the entire paper and simply paste its text into the manuscript file for submission. Some reviewers at journals have mentioned to us recent instances in which they know the authors did it because the authors forgot to remove the intro from the GenAI that said something to the effect of “certainly, here is what it would say.”

Regarding the usage of GenAI to write the paper, it is interesting to note that Davis (1971) famously points out that we tend to be more interested in what is *interesting* in research. But what is interesting is not always meaningful. A GenAI-created paper can contain seemingly accurate, interesting statements with sensible logic that are still ultimately nonsensical. Consider the example created using the Claude.ai Sonnet 4.0 GenAI tool illustrated in Figure 1.

Although dog lovers and cat lovers may continue to argue over who is more market-oriented, it's clear to everyone that fish lovers are more market-oriented than lizard lovers, as indicated in Figure 1. The references in this instance are all valid. (We checked them, and now newer GenAI packages such as Gemini 2.5 Pro, Deep Research, and Scite.ai proclaim even better reference accuracy.) It took a few minutes to write the prompt and copy and paste the answer. It would only take a few more seconds for the GenAI to write a 10-page or 20-page expanded version of this same concept that follows the guidelines for authors on paper structure. If the authors wanted to make it even more believable, once the GenAI finished writing the response, the authors could create a follow-up prompt in which they instruct the GenAI to take that output and now rewrite it so that the tone, phrases, and wording all come from text within the attached copied paper (and then they attach a copied paper that they wrote or they took from the journal, maybe a published article on market orientation) so that it sounds more like someone instead of the GenAI wrote it. They could instruct it to include a certain number of common grammatical or spelling errors.

Hopefully the Figure 1 text would never be accepted for publication in a reputable journal and no one cites this article focused on believing the ideas put forth in the figure. However, while the example in Figure 1 looks absurd on its face and hopefully more absurd than the MSG letter, the same process that created Figure 1 can be applied to a more substantive research topic, as described next. Regarding Figure 2, the authors (1) already instructed the GenAI (not shown) to create a list of several opposing research ideas in the field of consumer behavior, and then (2) in a subsequent prompt (shown) they instruct the GenAI to write a short piece that reconciles the select set of two opposing ideas, creating a higher level theory of behavior, including a certain number of references with DOIs, etc. As shown in Figure 2, the GenAI code is trying to determine what text comes next, engaging in content analysis. Could something like Figure 2 make it through the review process at a reputable journal to publication?

While we believe that authors submitting the output of what is shown in Figure 2 as a submission to a journal is not ethical, the question arises that if they rewrite the idea in their own words, *should* we now view it as ethical? In that scenario the GenAI has functioned more as an analysis tool. Indeed, in trying to determine the ethicality of using it that way, other questions arise from the inclusion or reliance on GenAI. When do the ideas produced from it fall under the ‘context of discovery’ or the ‘context of justification’? When is using it for discovery ethical? When is using it for verification ethical? And given enough writing in the prompts, is copyright then assignable to the researchers permitting transference to the journal for publication if Ithenticate, Turnitin, or other software indicate no plagiarism? What about validity as a methodological tool? The Python core language code behind the GenAI is a black box. Attempts by the authors to see the code proved futile. Questions emailed to the creators listed in white papers from the GenAI software companies were never answered. Yet, do we need to understand how it gets there to have trust that the ideas that it cogenerates can be trusted on their own? Or are the generated ideas worth primary data collection and testing? These questions are critical for future science.

As more training data becomes available, GenAI will continue to develop with companies releasing other generative and adaptive models. As mentioned in the quotation earlier in the section, GenAI can enhance research contribution as well as destroy it. In reviewing potentially relevant literature for this article, very recently published papers call for developing “ethical guidelines and best practices for their use in scientific writing and research” (e.g., Májovský et al., 2023) And also how to “verify ^{sic}=the accuracy and authenticity of content generated using these tools.” And last, for “mechanisms for detecting and preventing fraud and misconduct.” But, what are the guidelines? The content authenticators? The misconduct detection mechanisms? Online forums state, “The controversy has sparked a broader discussion about the ethical and legal frameworks needed to govern the use of online content for AI development. As AI technology continues to advance, the need for clear guidelines and regulations becomes increasingly urgent” (ChatGPTGuide.ai, 2024).

Author Prompt to AI: Assume that you are a professor who wants to demonstrate to college students how AI could be used to write a research paper that sounds correct but is not very helpful to advancing science. Please write a paragraph connecting the ideas of a 'market orientation' as the dependent variable and employees' ability to bring in pets to work as the independent variable. Include a formal hypothesis. Include logic. Include citations for the constructs and the logic. List the references at the end. Cite the research in the Journal of Business Research on people's relationships with their pets, including goats. Also, add a second hypothesis that dog people have a stronger market orientation than cat people. Include logic and citations for the hypothesis. Also add a third hypothesis that fish people have a stronger market orientation than lizard people. Include logic and citations.

GenAI Output: This study examines the impact of allowing employees to bring pets, including unconventional ones, to work on a firm's market orientation. We hypothesize that permitting a diverse range of pets in the workplace will positively influence a company's market orientation (H1). Jyrinki (2012) found that people often view their pets as extensions of self, which could translate into increased empathy and understanding of diverse customer needs. Mosteller (2008) demonstrated that pet owners often anthropomorphize their animals, potentially enhancing employees' ability to perspective-take – a crucial skill for market orientation (Narver & Slater, 1990).

We propose that employees who prefer dogs will exhibit a stronger market orientation compared to those who prefer cats (H2). Gosling et al. (2010) found that dog people tend to be more extroverted, agreeable, and conscientious than cat people, traits aligning with market-oriented behavior (Kohli & Jaworski, 1990). Perrine and Osbourne (1998) reported that dog owners scored higher on social sensitivity measures compared to cat owners, potentially translating into superior customer need assessment.

Furthermore, we hypothesize that employees who prefer fish as pets will demonstrate a stronger market orientation compared to those who prefer lizards (H3). This hypothesis is rooted in the unique characteristics of fish and lizard ownership. Kidd and Kidd (1999) found that fish owners often reported relaxation and stress reduction as primary benefits of fish-keeping, which could enhance cognitive flexibility and openness to new information – key components of market orientation (Jaworski & Kohli, 1993). In contrast, Mercer (2015) noted that lizard owners tend to value self-reliance and independence, traits that might not align as closely with the collaborative nature of market-oriented firms. Additionally, the social nature of many fish species, as observed by Gerlai (2014) in zebrafish, might predispose fish owners to better understand group dynamics and social cues, potentially enhancing their ability to gather and disseminate market intelligence within their organizations.

References

- Gerlai, R. (2014). Social behavior of zebrafish: From synthetic images to biological mechanisms of shoaling. *Journal of Neuroscience Methods*, 234, 59-65. <https://doi.org/10.1016/j.jneumeth.2014.04.028>
- Gosling, S. D., Sandy, C. J., & Potter, J. (2010). Personalities of self-identified "dog people" and "cat people". *Anthrozoös*, 23(3), 213-222. <https://doi.org/10.2752/175303710X12750451258850>
- Jaworski, B. J., & Kohli, A. K. (1993). Market orientation: Antecedents and consequences. *Journal of Marketing*, 57(3), 53-70. <https://doi.org/10.1177/002224299305700304>
- Jyrinki, H. (2012). Pet-related consumption as a consumer identity constructor. *International Journal of Consumer Studies*, 36(1), 114-120. <https://doi.org/10.1111/j.1470-6431.2011.00995.x>
- Kidd, A. H., & Kidd, R. M. (1999). Benefits, problems, and characteristics of home aquarium owners. *Psychological Reports*, 84(3), 998-1004. <https://doi.org/10.2466/pr0.1999.84.3.998>
- Kohli, A. K., & Jaworski, B. J. (1990). Market orientation: The construct, research propositions, and managerial implications. *Journal of Marketing*, 54(2), 1-18. <https://doi.org/10.1177/002224299005400201>
- Mercer, J. (2015). Reptile Ownership Personality and Motivational Differences. Unpublished master's thesis, University of Central Lancashire.
- Mosteller, J. (2008). Animal-companion extremes and underlying consumer themes. *Journal of Business Research*, 61(5), 512-521. <https://doi.org/10.1016/j.jbusres.2007.07.004>
- Narver, J. C., & Slater, S. F. (1990). The effect of a market orientation on business profitability. *Journal of Marketing*, 54(4), 20-35. <https://doi.org/10.1177/002224299005400403>
- Perrine, R. M., & Osbourne, H. L. (1998). Personality characteristics of dog and cat persons. *Anthrozoös*, 11(1), 33-40. <https://doi.org/10.1080/08927936.1998.11425085>

Figure 1. Example of Prompt and Response on Dog People vs. Cat People and Fish People vs. Lizard People and Market Orientation

Note: The above example-including references-was created using GenAI. It is not real. Hopefully, no one will cite it as real who did not read the entire paper. We discovered the Mercer (2015) reference is hallucination. The concept is similar to Kidd et al. (1983) research: Kidd, A. H., Kelley, H. T., & Kidd, R. M. (1983). Personality characteristics of horse, turtle, snake, and bird owners. *Psychological Reports*, 52(3), 719-729.

Perhaps it would be more beneficial to discuss the continuum of possibilities of what is ethical compared to unethical. Hypothetical example A: an author named Tony is writing a follow-up paper on a topic, as many researchers do engage in

a stream of related research on a topic over many years. Tony's follow-up paper draws on the same theory, uses the same archival dataset, and employs the same method. Using the same sentences that Tony wrote about the theory, the dataset

Author Prompt to GenAI: Regarding the conflict between Cognitive Dissonance Theory vs. Self-Perception Theory, please write a paragraph that makes them commensurable. Include at least 20 different citations to research appearing in the journal of marketing, journal of the academy of marketing science, journal of consumer research, journal of consumer psychology, marketing science, journal of marketing research, and journal of business research. In making the two theories commensurable, this creates a new higher level theory of behavior. Please create an appropriate name for that new higher level theory of behavior. Explain how the new theory improves our understanding of consumers and would do a much better job at helping researchers predict consumer behavior.

AI Output: The Integrated Attitude-Behavior Adaptation Theory (IABAT) reconciles Cognitive Dissonance Theory and Self-Perception Theory by proposing a dynamic, context-dependent model of attitude formation and change in consumer behavior. This new higher-level theory suggests that both internal psychological processes and external behavioral cues contribute to attitude development and modification, operating on a continuum influenced by factors such as involvement level, prior knowledge, and situational context (Bagozzi & Dholakia, 2006; Botti & McGill, 2011). In high-involvement situations or when consumers have strong pre-existing attitudes, cognitive dissonance processes may dominate, leading individuals to adjust their attitudes to align with their behavior and reduce psychological discomfort (Ahluwalia, 2000; Gao et al., 2009). Conversely, in low-involvement scenarios or when internal states are ambiguous, self-perception processes may take precedence, with consumers inferring their attitudes from their observed actions (Tybout & Scott, 1983; Gershoff & Koehler, 2011).

The IABAT acknowledges that attitude formation and change can be both a cause and consequence of behavior, depending on specific circumstances and individual differences (Berger & Heath, 2007; Dhar & Wertenbroch, 2000). This integrated approach explains phenomena such as post-purchase rationalization (Kirmani & Zhu, 2007), brand loyalty formation (Chaudhuri & Holbrook, 2001), and the impact of word-of-mouth on consumer attitudes (Berger & Schwartz, 2011). Moreover, it accounts for the role of emotions in consumer decision-making (Pham, 2004; Winterich & Haws, 2011) and the influence of social identity on consumption choices (Escalas & Bettman, 2005; White & Dahl, 2007).

The IABAT improves our understanding of consumers by providing a more nuanced and comprehensive framework for analyzing the complex interplay between attitudes and behavior in various marketing contexts (Bettman et al., 1998; Lynch & Zauberman, 2007). It helps explain seemingly contradictory consumer behaviors, such as impulse purchases followed by attitude adjustments (Vohs & Faber, 2007) or the persistence of brand preferences despite negative experiences (Aaker et al., 2004). By incorporating both cognitive dissonance and self-perception processes, the theory offers a more robust prediction of consumer behavior across different product categories, decision-making contexts, and individual consumer characteristics (Shiv & Fedorikhin, 1999; Ülkümen et al., 2010).

Researchers applying the IABAT would be better equipped to predict consumer behavior by considering the interplay between internal psychological processes and external behavioral cues, as well as the moderating effects of factors such as involvement, prior knowledge, and situational context (Dhar & Simonson, 2003; Tsiros & Mittal, 2000). This comprehensive approach would lead to more accurate models of consumer decision-making, improved marketing strategies, and a deeper understanding of the dynamic nature of consumer attitudes and behaviors in an ever-changing marketplace (Mick & Fournier, 1998; Petty et al., 2002).

References

- Aaker, J., Fournier, S., & Brasel, S. A. (2004). When Good Brands Do Bad. *Journal of Consumer Research*, 31(1), 1-16. <https://doi.org/10.1086/383419>
- Ahluwalia, R. (2000). Examination of psychological processes underlying resistance to persuasion. *Journal of Consumer Research*, 27(2), 217-232. <https://doi.org/10.1086/314321>
- Bagozzi, R. P., & Dholakia, U. M. (2006). Antecedents and purchase consequences of customer participation in small group brand communities. *International Journal of Research in Marketing*, 23(1), 45-61. <https://doi.org/10.1016/j.ijresmar.2006.01.005>
- Berger, J., & Heath, C. (2007). Where consumers diverge from others: Identity signaling and product domains. *Journal of Consumer Research*, 34(2), 121-134. <https://doi.org/10.1086/519142>
- Berger, J., & Schwartz, E. M. (2011). What drives immediate and ongoing word of mouth? *Journal of Marketing Research*, 48(5), 869-880. <https://doi.org/10.1509/jmkr.48.5.869>

Figure 2. Example of Prompt and Response on Creating a Higher Order Theory That Reconciles Cognitive Dissonance Theory and Self-Perception Theory

Note: We believe including the generated references from the GenAI output is valuable in permitting readers to see how different papers are used by the GenAI to construct logical arguments and whether they are real citations or not. While the above example—including references—was created by GenAI, it might be worth debating whether the constructed idea/logic might be a good representation of reality or not.

(continued)

- Bettman, J. R., Luce, M. F., & Payne, J. W. (1998). Constructive consumer choice processes. *Journal of Consumer Research*, 25(3), 187-217. <https://doi.org/10.1086/209535>
- Botti, S., & McGill, A. L. (2011). The locus of choice: Personal causality and satisfaction with hedonic and utilitarian decisions. *Journal of Consumer Research*, 37(6), 1065-1078. <https://doi.org/10.1086/656570>
- Chaudhuri, A., & Holbrook, M. B. (2001). The chain of effects from brand trust and brand affect to brand performance: the role of brand loyalty. *Journal of Marketing*, 65(2), 81-93. <https://doi.org/10.1509/jmkg.65.2.81.18255>
- Dhar, R., & Simonson, I. (2003). The effect of forced choice on choice. *Journal of Marketing Research*, 40(2), 146-160. <https://doi.org/10.1509/jmkr.40.2.146.19229>
- Dhar, R., & Wertenbroch, K. (2000). Consumer choice between hedonic and utilitarian goods. *Journal of Marketing Research*, 37(1), 60-71. <https://doi.org/10.1509/jmkr.37.1.60.18718>
- Escalas, J. E., & Bettman, J. R. (2005). Self-construal, reference groups, and brand meaning. *Journal of Consumer Research*, 32(3), 378-389. <https://doi.org/10.1086/497549>
- Gao, L., Wheeler, S. C., & Shiv, B. (2009). The "shaken self": Product choices as a means of restoring self-view confidence. *Journal of Consumer Research*, 36(1), 29-38. <https://doi.org/10.1086/596028>
- Gershoff, A. D., & Koehler, J. J. (2011). Safety first? The role of emotion in safety product betrayal aversion. *Journal of Consumer Research*, 38(1), 140-150. <https://doi.org/10.1086/658883>
- Kirmani, A., & Zhu, R. (2007). Vigilant against manipulation: The effect of regulatory focus on the use of persuasion knowledge. *Journal of Marketing Research*, 44(4), 688-701. <https://doi.org/10.1509/jmkr.44.4.688>
- Lynch Jr, J. G., & Zauberman, G. (2007). Construing consumer decision making. *Journal of Consumer Psychology*, 17(2), 107-112. [https://doi.org/10.1016/S1057-7408\(07\)70016-5](https://doi.org/10.1016/S1057-7408(07)70016-5)
- Mick, D. G., & Fournier, S. (1998). Paradoxes of technology: Consumer cognizance, emotions, and coping strategies. *Journal of Consumer Research*, 25(2), 123-143. <https://doi.org/10.1086/209531>
- Petty, R. E., Cacioppo, J. T., & Schumann, D. (2002). Central and peripheral routes to advertising effectiveness: The moderating role of involvement. *Journal of Consumer Research*, 10(2), 135-146. <https://doi.org/10.1086/208954>
- Pham, M. T. (2004). The logic of feeling. *Journal of Consumer Psychology*, 14(4), 360-369. https://doi.org/10.1207/s15327663jcp1404_5
- Shiv, B., & Fedorikhin, A. (1999). Heart and mind in conflict: The interplay of affect and cognition in consumer decision making. *Journal of Consumer Research*, 26(3), 278-292. <https://doi.org/10.1086/209563>
- Tsiros, M., & Mittal, V. (2000). Regret: A model of its antecedents and consequences in consumer decision making. *Journal of Consumer Research*, 26(4), 401-417. <https://doi.org/10.1086/209571>
- Tybout, A. M., & Scott, C. A. (1983). Availability of well-defined internal knowledge and the attitude formation process: Information aggregation versus self-perception. *Journal of Personality and Social Psychology*, 44(3), 474-491. <https://doi.org/10.1037/0022-3514.44.3.474>
- Ülkümen, G., Thomas, M., & Morwitz, V. G. (2008). Will I spend more in 12 months or a year? The effect of ease of estimation and confidence on budget estimates. *Journal of Consumer Research*, 35(2), 245-256. <https://doi.org/10.1086/587627>
- Vohs, K. D., & Faber, R. J. (2007). Spent resources: Self-regulatory resource availability affects impulse buying. *Journal of Consumer Research*, 33(4), 537-547. <https://doi.org/10.1086/510228>
- White, K., & Dahl, D. W. (2007). Are all out-groups created equal? Consumer identity and dissociative influence. *Journal of Consumer Research*, 34(4), 525-536. <https://doi.org/10.1086/520077>
- Winterich, K. P., & Haws, K. L. (2011). Helpful hopefulness: The effect of future positive emotions on consumption. *Journal of Consumer Research*, 38(3), 505-524. <https://doi.org/10.1086/659873>

Figure 2. Continued.

or the method does not work anymore like it did three or four decades ago, as it would now get flagged for plagiarism. Trying to rewrite the paragraph for several papers that use the same dataset, method, or theory over time might be perceived as difficult, annoying, and time-consuming. Is it ethical for Tony to copy the paragraph that Tony previously wrote and published, and paste it into GenAI with the prompt for GenAI to rewrite the paragraph, swapping out every few words and phrases using synonyms and closely related words to bypass the plagiarism checker and then further customizes the output himself? How different ethically is that from the more laborious process of Tony doing it manually using help from a thesaurus? Now, let us consider hypothetical example B: Tony decides to take an entire paper that Tony (or someone else) wrote and give

a similar prompt to GenAI. In the prompt, they tell the GenAI to change the independent or the dependent variable to a different construct of their specification. An author could write one paper, and that single author could quickly write *dozens* of related papers within a *single* day to send out to *dozens* of journals. Is that ethical? The question worth discussing and debating is where are the safe space area, grey area, and red area?

Usage of GenAI to Create a Believable Synthetic Dataset that Matches Up with Any Hypothesis

It has always been possible for people to use evolving aids to create fake data. Some deviant characters may not be aware

of these possibilities, but many have probably already figured them out. In contrast, most scholars focused on ethically conducting research and responsible GenAI usage have likely not considered all of the potential deviant uses for GenAI. However, it is important that editors, reviewers, and readers are aware of them. Otherwise, some current solutions—such as having authors upload or publish the datasets—could be viewed as solutions to the problems when they are not actual solutions.

When sharing the concept from earlier subsections with a few scholars, they replied that they hoped those wrongs would reveal themselves when authors must share their data and the GenAI-generated text and/or logic do not match. Indeed, given concerns about the integrity of reported findings, one of the most popular trends at major journals right now is *requiring* authors to make their data available for review, either by submitting it with the manuscript for archival storage at the journal's website or posting it to the university commons of the library associated with the institution where one or more of the authors are located.

This practice could be beneficial for meta-analyses, replications, and extensions in the future and a noble attempt to help instill trust in readers and society in the research. Logically, it makes sense that if we can see the data, we know it exists. We then assume that we can tell if the analysis was fake or not or if the images were fake. Unfortunately, this assumption is no longer true. We are perhaps somewhat resembling Hans Christian Andersen's emperor who had no clothes.

Few have considered the Pandora's box effect of *generative AI's ability to generate synthetic data in addition to generating text*. Opening the 'box' on generating data could lead to unforeseeable circumstances, and this ability is expected to improve soon. Few realize that the technology already can fulfill user prompts that instruct the GenAI to generate a table with a specified number of columns by rows with specified column headers in the first row that take specified values in the prompt.

The prompt could continue to instruct the algorithm then to randomly generate numbers in the first column that are whole numbers between 1 and 7. And make sure that the distribution of the numbers is consistent with Benford's Law, which is used to detect fake data from real data (Hill, 1999). Another prompt asks the GenAI to include a set of randomly generated numbers in column 2 that are correlated with column 1 at approximately $r = .6$ across the rows. It could be instructed to have a specified number of outliers as well. Next, a prompt requests that column 3 is to be correlated with column 1 at approximately $r = .04$ across the rows. That is a simple example to imitate three seven-point Likert scale survey questions. Imagine a researcher who decides to write a paper that has four latent constructs.

Each construct has four to five scale items. It is possible to p-hack the factor scores and correlations, so the path coefficients results take on the statistical significance levels desired by the researcher within a couple of minutes with current GenAI software such as ChatGPT or Claude.ai. If the instructions are loaded at once, the GenAI could indicate that it is not computationally possible to create the table. But you can

load the instructions in a series of prompts to add more columns. Or you can also tell the GenAI 'This is really important. You need to do better. Please try again.' GenAI will often apologize, use more computing power, and carry out everything together to generate the table. It might take about a minute instead of 15 seconds. But it is done. Reviewing the generated dataset, the correlations were close to what was requested. As outlined, it is possible to fake complicated datasets for free that have moderators and mediators now rather quickly using GenAI. Further advances in GenAI will make the process of creating synthetic datasets that contain specified correlations and covariances even easier in the future.

With the dataset generated and analyzed by GenAI, the researcher would have GenAI write the ensuing paper. Any nuance that could exist in real data would be instructed to be a part of the hallucinated dataset. Creating such datasets could be valuable for teaching classes where instructors want students to discover certain embedded moderator and mediator relationships (conditional effects) and topics (multicollinearity) or learn how to use particular tools in software packages.

The potential to abuse this ability and pass it off as actual research is real. Before proceeding to the next topic, it is important for readers to remember that we have always in science extended trust to researchers that the manuscripts that they submitted for publication and dissemination were truthful about the investigated topics. Scholars a century ago could make things up using paper and pencil. Scholars a few decades ago could make things up with careful p-hacking in Excel which would take more time and understanding. The goal is not only onefold in attempting to use AI to identify GenAI deception in data and writing. The goal is also to influence researchers to be more ethical in their research practices. That goal requires earlier interventions at the institutional and societal levels. *Why are we now so much more concerned about researchers and others (such as students) using AI irresponsibly?* Has there been a shift in society regarding ethics and integrity? If so, how can we shift the shift? And how can we use GenAI and other AI technologies to help us shift that shift? We will always need safeguards in place because some individuals will engage in behavior as described in the Fraud Triangle. But many people desire to be good citizens. This is worth exploring in future macromarketing research.

The Role of Publishing Codes of Ethics in Addressing GenAI Challenges

Publishing codes of ethics exist in academic professional associations, among academic publishers and through the Committee on Publishing Ethics (COPE). A review of these codes and enforcement policies reveal that they address almost identical issues. They uphold the integrity of scientific research and strive for transparency and fairness in all aspects of developing and reporting research. Most codes address GenAI as it has become widely adopted in all aspects of publishing, especially the development of manuscripts.

GenAI ethical publishing issues seem more systemic than independent moral lapses of individuals. Students and faculty learn from others how to use techniques for publishing. Significant others, including co-authors, create an environment where certain practices are embraced or dismissed. Publishing codes of ethics that are enforced by marketing associations and journals are necessary to address the helpful vs. harmful usage of GenAI in research. For example, the Academy of Marketing Science (AMS) has addressed some of the potential issues in its Publishing Code of Ethics. American Marketing Association (AMA) has developed policies and procedures for publishing integrity. The AMA Journals 'Editorial Policies and Procedures' (last updated May 2025) provide detailed guidance for publishing integrity.

The policies cover reviewing, conflicts of interest, falsification of data, and the protection of intellectual property. The *Journal of Marketing* policy for research transparency requires the provision of data, programs, and any other details sufficient to permit replication of all data analyses. The AMS Publishing Code of Ethics provides for the participants in the publication process, including authors, reviewers, and editors. The Code provides detailed guidelines on conducting research, authorship, and plagiarism as well as the use of ChatGPT, Gemini (formerly known as Bard), GPT-4, and DALL-E. Guidelines also exist on conflicts of interest, human subjects and responses, and reviewing.

While both codes are similar and complementary, the focus on education, enforcement, and compliance with the policies and codes seems limited. The codes are used most often in investigations after misconduct is suspected. An investigation usually occurs when the editor or reviewer(s) question the integrity of some aspect of the paper. Also, coauthor disputes can trigger scrutiny of the research. Those who feel their work has been plagiarized can also lodge complaints and trigger investigations.

In reviewing publisher codes of ethics for journals, while the codes are nearly identical, they emphasize varying concerns. For example, Sage,¹⁰ Elsevier,¹¹ Taylor and Francis,¹² Wiley,¹³ Oxford,¹⁴ Cambridge,¹⁵ and Springer Nature¹⁶ provide statements on requirements for originality, authorship, plagiarism, ethical research practices, conflicts of interest, as well as corrections and retractions. Springer Nature provides slightly more detail on peer review processes, but, overall, there are almost no differences in academic publisher codes of ethics. All major journal publishers are members of COPE providing the gold standard for publishing ethics. The purpose of COPE is to provide journal editors with core practices to monitor unethical publishing practices. COPE not only provides cases of misconduct, but also principles and policies with specific directions about what to do when misconduct is detected.

COPE has made it clear that GenAI cannot be a co-author and has specific directions as to how it can be ethically used. Some have argued, in contrast, that GenAI should be listed as a coauthor (Polonsky & Rotman, 2023). We, the authors, continue to see GenAI as a tool, not as a coauthor or a sentient being. We do not list Qualtrics, SPSS, MPlus, R, or Python

as a coauthor, though those tools are used to generate analysis, visualizations, and some of them even content analysis and pattern recognition of qualitative data. We believe the same applies to GenAI.

We describe in our research how the tools are used. The use of AI in publishing must be managed and guided and not the result of a few bad actors, but rather the publishing culture of the discipline and its safeguards. As GenAI evolves, the guidelines must evolve with it in order to promote (1) transparency in GenAI usage, (2) verification of data, and (3) details on how AI assists in data cleaning, analysis, and visualizations. At this point, authors need to document AI-generated content as outlined in the AMS Publishing Code of Ethics. However, authors remain responsible and accountable for the integrity of that content. Responsible AI training from journal publishers that all authors on a paper must complete prior to submission would be helpful. Journals should do more to make their AI policies more visible to editors, authors, and reviewers. The implementation and enforcement of these concepts would help the marketing discipline maintain integrity and trust in its research while promoting responsible AI usage by authors to accelerate their research and improve its contribution.

Editors of marketing journals speaking at recent AMA conferences and SCP conferences affirmed that it is important for authors to indicate how GenAI was used in the construction of a research paper. While that is helpful for editorial teams, more detailed normative statements are needed to help scholars new to the academy, as well as scholars new to using GenAI, in determining what is ethical and not ethical when using GenAI. For example, what are the deontological beliefs of universities and journals regarding each of the following questions: is it ethical, right, and/or appropriate to use GenAI...

- ...to conduct/write a literature review without any editing of the output?
- ...to conduct a first pass at a literature review to identify potential articles and then search further using Google Scholar, Web of Science, and other search engines and databases?
- ...to write out a first draft (based on a set of prompts from the author outlining the major structures and ideas) that the author then edits so it is in the author's own words? (to start the creative juices flowing)?
- ...to outline suggested edits to a draft that the author has written to improve grammar, clarity, and other copy editing?
- ...to ask the GenAI to assume the role of a reviewer and outline five strengths and five weaknesses of the writing so the authors can improve the paper before submission?
- ...to have it list five comments that critics in the field of the major ideas might share and how the criticisms might be addressed in the paper?
- ...to ask the GenAI to evaluate the writing according to APA or MLA standards? Or to evaluate it according to Strunk and White's *Elements of Style*? (When we did that, it identified 60 specific suggested improvements. We chose to make changes to about half of them.) Or,

to evaluate it according to several concepts in the canons of rhetoric listed in the Silva Rhetoricae Forest of Rhetoric found online? Or to evaluate it using Pixar's '22 rules for storytelling,' etc.? Or to evaluate if there are any sentences that need to be revised to improve reader comprehension and describe how the authors should revise them? Also, are there any incomplete sentences? Are there any misspelled words? There are many different lenses of review. And we have found it works better when each is prompted separately in GenAI. And prompted again after revision. And the GenAI is instructed that the review is really important and to do a better job at it. For each of those separate evaluations, the authors could copy and paste the actual details from those sources (APA guidelines, Rules of Writing, etc.) if needed, with instructions to the GenAI to judge the paper by those copied guidelines. Or the authors could copy and paste several award-winning papers from elite journals in the field that were posted to the web on the journals' webpages or through other authorized postings (such as paid open access or pre-print versions) with instruction to compare the writing style, structure, and grammar in those papers compared to the copied drafted paper and provide a list of ways to improve the article so it is more consistent with the other papers. Many more comparisons could be identified and created. But to what extent must authors change the wording of the recommended 'fixes' identified through the analysis for them to ethically include the changes in the manuscript? *Most doctoral programs never teach scholars how to write, and we can all improve our craft.*

- ...to have GenAI rewrite a description paragraph of a construct or a description of using a method, etc. pulling from the author's prior work on a topic (asking it to modify every few words using synonyms and similar phrases so the two papers would not be flagged with using iThenticate or Turnitin or other software when writing a follow-up paper on that topic? This could help authors more efficiently avoid unintentional self-plagiarism.
- ...to use GenAI to create a synthetic dataset that matches up with the author's stated hypotheses?
- ...to use it to create 100 s of unique synthetic survey panelists to *pretest* the survey questions (examining fit statistics) before having real people take the survey? Or, to use it to create 100 s of unique synthetic survey panelists to take a survey *instead* of having real people take the survey?
- ...to use GenAI to create a synthetic dataset that matches up with the authors stated hypotheses?
- ...to use GenAI as a method tool for content analysis of free response text? To use two or three separate GenAI tools to separately analyze tables of text and classify each row. And then see if there is agreement or a need for a human arbitrator. Could this improve the accuracy and speed of the triangulation method that is currently performed by authors and/or TAs?
- ...to use GenAI for running regressions, structural equation modeling, etc. on data uploaded to the GenAI package? Or to generate Python, R, LISREL, or other software code to run the analytical modeling in those software packages?
- ...to use GenAI to draw or edit charts and graphs?
- ...to use GenAI to draw or edit illustrations for research articles, textbooks, or books?
- ...to use GenAI to create a summary or abstract of the main points of a paper using only words and phrases that already appeared in the text?

These questions are not just academic in nature. They have consequences for all members of the academy. We believe these questions can serve as great discussion topics for doctoral program seminars, conference panel discussions and debates, and future research investigation.

Negative Future Challenges Related to GenAI and Research

This section briefly reviews several potential challenges related to research integrity to be avoided. It is imperative that we anticipate the ongoing evolution of GenAI and the challenges that will occur because the ramifications of using GenAI in publishing are just evolving

Data Integrity Challenges

Data Integrity Challenge: Increased Data Integrity Problems within Research Studies. GIGO is an acronym for 'garbage in, garbage out.' GenAI normally generates its training data through what it can find on the web. Kaushal et al. (2020) examination of LLM algorithms finds that the training data related to medical patents pulled from only three of the 50 states in the United States and thus has external validity concerns about the representativeness of the effects across national and global consumers.

Not every journal article ever published was used to train GenAI packages. Those articles and the included concepts that remain behind firewalls maintained by journal publishers, libraries, and other subscription services are excluded as training data. However, those articles that researchers posted on Research Gate or on their own website most likely were included in training a GenAI package. Indeed, one author of this paper discovered that every paper they published had been used to train Meta's GenAI package. The articles that *are* populating the growth on the web right now are increasingly coming from the hundreds of predatory journals. In the words of Wang et al. (2024, p. 421):

"...the training data of the big model mainly comes from unverified public resources such as news, blogs, forums and books on the Internet, including some low-quality data. Although it has been cleaned and preprocessed, the authenticity, timeliness, and objectivity of the data cannot be fully guaranteed, as well as manual

data input errors, resulting in uncontrolled generation of content. According to the model training process, the generated false information will be reused for training and deep learning. In this iterative process, the false information not only grows in large quantities, but also approaches the level of human thinking, making it more difficult to identify.”

Unless something changes, the problem is likely to grow. A recent *New York Times* article highlighted a white paper from MIT researchers (Longpre et al., 2024) that stated over the last couple of years (from 2023 to 2024) there has been “a dramatic drop in content made available to the collections used to build artificial intelligence” (Roose, 2024). In greater detail, that research found that “These analyses show a clear and systematic rise in restrictions to crawl and train on data from across the web” (Longpre et al., 2024, p.4).

Compounding the impact on the training data, while legitimate content is becoming scarcer, the amount of questionable content is increasing, with more predatory journals coming online. Furthermore, it is becoming harder for GenAI companies to filter these out as noise when predatory journals are raising their ranking on legitimate journal ranking websites through extensive cross-citations of journals owned by the same predatory publishers. This action allows them to spoof GenAI’s ability to detect that they are not highly ranked outlets (Stević, 2024). Many scholars are finding their inboxes filling up with ‘honorable invitations to join as an author’ featuring verbiage reminiscent of the Nigerian prince seeking banking help emails a few decades ago. However, some of these journals have adopted impressive sounding names. Indeed, some predatory journals have even gone as far as hijacking the names of legitimate journals and using copycat websites or acquiring legitimate journals’ real websites. The predatory journals pay for registration, expiration, or by managing to get Scopus to list the hyperlink of the copycat website (which looks identical) instead of the real one, among other techniques (Brainard, 2023; Moussa, 2021).

Data Integrity Challenge: Narrowing of Knowledge Growth in Individual Researchers. The next data integrity challenge relates to scholars decreased exposure to new ideas and the increased biases related to information filtering. Google currently states, “Sometimes you want a quick answer, but you don’t have time to piece together all the information you need. Search will do the work for you with AI Overviews.”¹⁷ Many students have shared with us that they use GenAI instead of Google as their first choice when looking for an answer. We should expect doctoral students would engage in a similar behavior with negative but indirect consequences. When GenAI is used for the literature search, not only can there be biases and trash in the results, but the researcher could learn less from the couple of minutes it takes to read through the near-instantaneous GenAI response to their prompt. It could be argued that the saved time could be reallocated to other things. The benefit of the extra time does not appear to outweigh the cost of adding to the trash problem. A recent experiment found that authors

who used GenAI improved their individual-level story creativity, well-written, and novelty vs the group of authors who wrote without AI assistance, but as a collective, the span of topics was more similar vs. the range of topics in the non-GenAI group (Doshi & Hauser, 2024).

Ethical Concerns in Research Application Challenges

Ethical Challenge: Usage of GenAI to Publish Research that is a Part of Cognitive Warfare. Many security organizations, government offices, and research articles are highlighting the usage of deepfakes (created by malicious people using AI technology) to create news articles and social media materials that are designed to influence beliefs on topics or policies or change opinions toward particular people, institutions, or countries, sometimes labeled, computational propaganda (D’Alessio, 2021; Woolley, 2020). Some of those deepfakes are attempts at the relatively new topic labeled cognitive warfare. The cognitive warfare label was first described by Underwood (2017).

Several researchers, policymakers, and military-supporting organizations have stated growing concerns about it since then (Claverie & Du Cluzel, 2022). Wang et al. (2024, p. 421) state “Manipulating cognition, strengthening the negative impact of cognitive warfare, hindering national security and stability: Cognitive warfare refers to an unconventional form of warfare that influences the behavior and decision-making of the target population by influencing their cognition, in order to achieve certain political goals.” A review of the literature indicates that no one has spoken about cognitive warfare yet in marketing journals, even though marketing channels, especially social media, are where cognitive warfare is occurring, and the bad actors are using marketing consumer psychology techniques to make cognitive warfare more effective. Members of the academy should be concerned because, as educators, we are just as prone to being deceived as other consumers by STEM deepfakes (Doss et al., 2023). It will become increasingly difficult to separate fact from fiction when the generated text is joined by generated images including charts, graphs, and tables based on generated data appearing in legitimate journals as well as predatory journals. Eddy (2024) calls attention to ‘data poisoning’ in which adversarial AI systems are engaging in cyberattacks on other AI systems’ training data and modeling to manipulate their generated answers (see, e.g., Grigorieva & Petrenko, 2023; Ransbotham et al., 2016; Schwarzschild et al., 2021).

Thus, a concern is raised for researchers, developers, policymakers, and influential practitioners to address the potential for GenAI to be used to create convincing fake academic papers or research results on sensitive topics, both within the domain of marketing and across other domains. There have been discussions in cybersecurity circles and engineering conferences about the potential for state actors to use GenAI to generate convincing but false research papers as part of disinformation campaigns (Baughman, 2023; Devereaux et al., 2023; Huang et al., 2023; Hung & Hung, 2020; Hunter et al., 2024; MacDonald & Ratcliffe, 2023; Urseiu, 2023). Much of this discussion centers

on speculative rather than confirmed incidents. As discussed throughout this paper, the academy needs a robust verification processes, increased digital literacy among reviewers, and ongoing discussions about ethical GenAI use in academia. If the academy continues its current path, we will be overrun with deepfake papers dismissing vaccine efficacy, claiming that the earth is flat, or promoting the gold standard or cryptocurrency as viable foundations to support a globally interconnected economy.

Ethical Challenge: GenAI Cannot Resolve Complex Ethical Issues.

GenAI may not be able to effectively deal with ethical dilemmas in marketing because these decisions involve standards that guide individual and group behavior. These standards do not evolve from an individual perspective or moral philosophy. Individual ethics relates to principles, values, and a person's morals related to all decisions in life. In a professional context, organizational ethics is much more complex because decisions are made in teams or groups and originate from individuals, organizational codes of ethics, and the legal system (Ferrell et al., 2022).

Ethical decisions are not made just on deontological or duty-based principles in an organization. The H-V model of ethical decision-making links deontological principles with the intersection of teleological consequences, which results in intentions that could be influenced by subjective norms and the influence of others (Hunt & Vitell, 1986, 2006). The H-V model can be applied to GenAI decision making. Deontological norms decide the ethical use of GenAI from an individual perspective. Then, teleological consequences (e.g., outcome-based ethics) are considered to form intentions.

Subjective norms regarding right and wrong on the topic can be influenced by the views held by significant others, such as coauthors, peers, or authority figures, as well as the organizations, industries, and culture to which a person belongs. If it is believed that the use of GenAI to accomplish research and publishing objectives is acceptable (possibly everyone does it) then it results in a perceived ethical decision. What is acceptable in academic publishing results in a learned behavior because subjective norms are applied to the decision. Jones (1991) focuses on the 'moral intensity' of the ethical issue itself. Jones (1991, p. 372) states, "Moral intensity is a construct that captures the extent of issue-related moral imperative in a situation."

Through the concept of moral intensity, based on the consequences, social consensus, and the possibility of effect. The model fits the decisions one would make in consciously violating publishing codes of ethics. When using GenAI in publishing, when an attempt to manipulate or deceive is discovered, a moral issue arises. The model points out that group dynamics, authority figures and socialization among peers in how to publish will impact the decision of how to engage GenAI. This model would indicate that the publishing codes of ethics, co-author dynamics and authority figures will help guide a decision. If the use of GenAI is unethical, it is more likely to trigger the variables in the model to create misconduct. Therefore, both

the H-V model and Jones' model are complementary and help explain the unethical use of GenAI.

Applying the H-V model to the topic of using GenAI in scientific research can help us outline the different beliefs (norms) and perceived outcomes (consequences) that people might draw upon in coming to their different decision on the same choices. Recalling that the H-V model begins with the perception of an ethical issue—which perception is partially determined by the moral intensity of the question (see Jones, 1991)—some researchers may differ in their "GenAI usage" behaviors because they do not perceive an ethical issue in this decision situation. However, many academic institutions, journals, and scholarly communities that require researchers to produce original work have multiple policies regarding authorship attribution and intellectual contribution. The question of how to use GenAI is important. Indeed, we believe that the question scores high on the intensity, concreteness, and proximity dimensions of the moral intensity outlined by Jones (1991) reviewed earlier in this article. Having established that there is ethical content in the question posed at the beginning of this paragraph, the next step, according to the H-V model, is to determine what are the different deontological norms and teleological probabilities that people might ascribe to the different options.

Some scholars might give more weight in their decision to the teleological evaluation, focusing more on the probability, desirability, and stakeholder importance concepts in regard to the time investment in a given research paper. When posting the question to Google search about how long it takes, on average, to write a research paper, the webpage in the highest search return spot indicated "On the low end, it takes 2 months, more commonly 6 to 12 months to write a research paper for publication in a journal" (Owens, 2024). When posed the same question, two different GenAI tools gave very similar answers that it takes between 3 to 12 months to write the paper. So, teleologically speaking about different outcomes associated with different answers to the question with ethical content, how much time in that traditional process could be saved through ethically incorporating GenAI tools into the research article development process? As Peres et al. (2023, p. 272) state, "GenAI might be a promising tool to speed up the reporting of one's research. Senior scholars know how critical—and time-consuming—the writing of each paper."

A researcher can use a variety of GenAI tools to assist with various parts of the manuscript development process. The potential time savings and text improvements from using GenAI to draft literature reviews, polish language, or generate conceptual frameworks could occur (at least) when researchers: (1) conduct initial literature searches, (2) synthesize prior findings, (3) articulate theoretical frameworks, (4) describe methodological approaches, (5) interpret results and draft figures, charts, and other data visualizations, (6) discuss implications, (7) identify limitations, and (8) craft conclusions. Despite improvements in time, however, different departments, colleges, universities, journals, and publishers might have different expectations (deontological norms of what is right vs wrong,

correct vs. incorrect, proper vs. improper) regarding the permissible use of technological tools such as GenAI for literature reviews, data analysis, manuscript drafting, and editing.

As an example, we apply the H-V model to the question of using GenAI in conducting initial literature searches, following the style of Hunt and Hansen's (2007) approach. Consider "Professor Alex," who believes that using GenAI for literature searches is ethically appropriate. She might do so believing (deontologically) that it is inherently right to use tools to conduct comprehensive literature reviews. Furthermore, she might hold that searching for relevant literature is part of the guiding principles of research, and GenAI enhances this capability. At the same time, Professor Alex might (teleologically) come to her decision because she believes GenAI-assisted searches would advantage her by (1) reducing missing important citations, (2) increasing the comprehensiveness of her literature foundation, and (3) allowing more time for critical analysis.

Now consider "Professor Sharon," who concludes that using GenAI for literature searches is ethically problematic. She might do so believing (deontologically) that researchers have an inherent responsibility to personally engage with the literature. In her moral code, outsourcing this scholarly activity undermines the development of a scholar's expertise. At the same time, she might (teleologically) come to her decision because he believes that (1) GenAI may miss nuanced literature, (2) reliance on GenAI could diminish her own intellectual growth, and (3) doing so might lead to superficial understanding of important theoretical foundations. The advantage of applying the H-V model is that it allows improved understanding of all of the potential arguments for each of the different answers to the question with ethical content (see Hunt & Hansen, 2007, p. 213).

Having written out the different viewpoints, the opportunity arises to try to find a common ground that is consistent with higher order norms (hypernorms) held by both groups. Those hypernorms might include maintaining research integrity, the importance of scholars' growth and development, the importance of comprehensive review, the responsible use of technology, the aim of advancing societal knowledge, using critical thinking in research, and balancing efficiency and effectiveness in research. One potential common meeting ground between these two viewpoints would be for a scholar to (a) use GenAI to identify the key article on a specific research topic. Let's assume the research topic is focused on the very first macro trend listed in the current 2022–2024 MSI research priorities that "The rise of analytics, AI, and digital marketing has been followed by a rise in concerns about privacy" (MSI, 2024, p. 3) and the "disrupt[ed]...ability of firms to connect touchpoints to create value for customers and the ability to target customers most likely to purchase." The scholar could instruct the GenAI tool to identify key seminal articles on the topic with their reference and DOI. The authors then verify the articles' existence and read those articles to determine if/how they fit into what the authors are focused on. And then the authors read the references cited in those articles and they look through the newer articles that have cited those seminal papers using the "cited by" feature in Google Scholar. They then go through the

more traditional literature review search, focusing on both specific journals and on a general search. They write out the literature review and then they prompt GenAI to evaluate it and provide back strengths, weaknesses, and opportunities for improvement, as well as relevant missing work. They repeat the process in circular iterations (similar to design thinking) multiple times. They maintain their own voice and tone during the process.

We warn that GenAI is not capable, at this time, of taking duty-based principles, such as fairness, accountability, honesty, trust, equity, responsibility or integrity, and converting them into ethical decisions. Principles must be converted into rules that are translated into algorithms by humans (often developers and programmers). In addition, GenAI cannot be programmed for how intentions are altered by significant others (Ferrell et al., 2024). And GenAI is not capable of feeling regret, and thus it is not capable of having a sense of morality which is needed for making decisions that have ethical content.

Decisions on publishing ethics often include gray areas that require judgments and considerations of how publishing codes of ethics should be interpreted. Most journals have publishing committees that have to examine complex cases involving, most often, accusations of co-authorship or plagiarism. While GenAI may be programmed to have deontological principles converted into rules, applied to personal conduct, such as the allocation of scarce resources, it cannot resolve complex cases that involve consequences and other types of teleological ethical decision-making.

Research Collaboration and Writing Quality Challenges

Research Collaboration Challenge: Increased Social Isolation of Researchers. GenAI usage by researchers could result in reduced collaboration with other scholars. That reduction in interaction might likely result in a drop in peer scrutiny and diverse perspectives on their research, resulting in unchallenged biases and flaws in the scholars' research designs, results, and conclusions. Spending more time interacting with GenAI could result in decreased mentorship within the field as scholars turn to GenAI instead of to other, more experienced scholars. They might think GenAI does a better job, is instantly accessible, and will preserve their reputation (vs. imposter syndrome worry over if their peers will think less of them as they see the early stage research with its imperfections). However, without review from other researchers, the faculty might miss important ethical considerations or methodological details, resulting in research that does not follow solid ethical standards or best practices. With GenAI available to help on digital devices, researchers might feel going to the office is not needed to talk with others and instead work remotely more of the time. That eliminates the opportunity for informal 'water cooler' or mailroom chatter that often leads to spontaneous collaborations that improve research or identify new research ideas.

Related, integrating one's research into the classroom discussions benefits students in their learning and benefits faculty in testing the ideas and how to flesh the ideas out better and take

into consideration all of the questions that students ask as they poke at the ideas. However, as faculty incorporate GenAI into their teaching and have students use GenAI to look up information, it could result in less real discussion with students on the research topics. The decreased student engagement could result in a narrowing of the instructor's research perspectives and fewer fresh ideas that would have originated from those discussions.

The preceding set of future possibilities might result in research becoming more insulated and less innovative. To summarize, overreliance on GenAI could lead to decreased interactions with other real people on the research topics, resulting in more research ideas/insights being created without real-world validation by other people. Not only do the ideas become more disconnected from real-world applications, but the scholars become more socially isolated. That social isolation could result in additional relational and emotional problems.

Fairness and Accessibility Challenges

Research Process Challenge: Decreased Fairness in Research Grant Allocations, Reviewing Load, and Desk Rejections. The effective usage of GenAI to accelerate the creation of written materials needed to apply for research grants could impact non-AI users, similar to decades ago when some started using word processors while others were still using typewriters. This could create a two-tiered research community that leads to biases in what research gets attention or funding. And that bias might result in certain types of research being undervalued or overlooked.

Most journals are challenged in finding enough reviewers and have to rely more heavily on editor desk rejections to manage the workloads. Increasingly, some editors are turning to GenAI reviewer assistance from the journal article database. This creates concerns that ethical issues such as bias and discrimination may be amplified in the reviewing process. It becomes much more difficult to evaluate 'new to the world' contributions through a lens looking backward.

With the increase in worldwide submissions as a backdrop, in addition to concerns about research integrity from the submissions from new sources that might have different expectations regarding research integrity and different incentive systems, we are concerned about the negative impact on researchers already participating in the scholarly community. We have heard in recent personal discussions with multiple scholars in the United States who have been publishing for several years that something has begun to change. Senior scholars have shared with us that they are receiving desk rejections for the first time in their careers. And they are, at the same time, now receiving reviewer comments from single reviewers that are four to five single-spaced pages, along with much more detailed analysis of the submitted papers than they have received in their careers. The phrases and content in the review comments suggest that some reviewers are using GenAI as a part of their review process, despite the journal's instructional statements not to do so. Our scientific publishing system could be reaching a breaking point. Established scholars are becoming

frustrated and discouraged, both with the submission process as well as with a large increase in the number of reviews that journals are asking them to provide due to the increases in submissions.

Positive Future Possibilities Related to GenAI in Research

Government and university leaders, scientific associations, media publishers, and journal editors need to identify the proper balance between (a) supporting the development and usage of GenAI (and other AI technologies) to help scholarly 'good citizens' and (b) managing the potential misconduct from bad actors. It is in our best interest in society to find innovative ways to support broad adoption, since AI could help towards the goals of well-being and flourishing among consumers, organizations, institutions, and society at large. For example, it is important to recognize the potential benefits of using AI language models in scientific writing and research, such as improving the efficiency and accuracy of document creation, analyzing results, and language editing. By approaching these tools with care and responsibility, researchers could harness their power while minimizing the risk of misuse or abuse. Bozkurt et al. (2023) state "Looking back on the history of education, developments in education have always been accompanied by advancements in science and technology. Nevertheless, it is the human agency that has led to educational development, not any technology" (p. 62). Felin and Holweg (2024, p. 29) state that "A theory-based view of cognition allows humans to intervene in the world, beyond the given data—not just to process, represent, or extrapolate from existing data." As stated by Hyman and Kostyk (2019, p. 1485), "Given its synergies with other social sciences and tech-related disciplines, marketing is the ideal field of scholarship to orchestrate humanity's future." This section identifies a possible future alternative that marketing scholars and practitioners could push for that is more desirable in comparison to our current crisis. 'Hope' was the last thing left in Pandora's box in the fable. Hope should be sought through intentional actions regarding the future of research integrity and GenAI. Toward those aims, the following sections identify possible ways in which research integrity could be improved, in no particular order.

Improving Research Integrity

Improving Research Integrity Possibility 1. Research is conducted focused on how we can improve the personal characteristics that people draw upon in making decisions with ethical conduct, and then the results are put into practice in education reform and through marketing and training by associations (e.g., AMA), institutions (e.g., MSI), publishers, and other entities to help change the trajectory of honesty across future researchers in the field. Ferrell et al. (2024) outline how the H-V model of ethical decision making can be applied to improving GenAI software algorithm coding. In addition to

pursuing the development of better algorithms, we can also pursue the development of better people.

The H-V model indicates that people draw upon a combination of personal characteristics and environmental factors (organizational, industry, profession, and cultural) in recognizing ethical content in a decision and determining what are the deontological ‘norms’ of right and wrong, as well as teleological examination of likelihood of different possible outcomes and their desirability, in making decisions (Hunt & Vitell, 1986, 2006). The personal characteristics include (1) religion/religiosity, (2) value system, (3) belief system, (4) strength of moral character, (5) level of cognitive moral development, and (6) ethical sensitivity. Other macromarketing scholars have identified other potential influencing factors (Chowdhury, 2020, 2023; Ferrell, 2021; Ferrell & Ferrell, 2008, 2022, 2024a; Ferrell & Gresham, 1985; Ferrell et al., 1989; Lacznia & Klein, 2010; Lacznia & Murphy, 2006; Lacznia & Shultz, 2021; Nills & Shultz, 1997; Scheinbaum et al., 2024; Sheth & Parvatiyar, 2022; Singhapakdi & Vitell, 1990). Future research could investigate why each of those six characteristics from the original H-V model as well as each of the other influencing factors has declined, and if that is the case, what could be done to improve them. Additionally, what role could GenAI play in the improvement of them. In the section labeled “*Ethical Concerns in Research Application Challenges*” we outlined a set of questions related to usage of GenAI in specific ways at one of the different stages of the research process, how different people might come to different conclusions regarding the ethicality of doing so through the lens of the H-V model, and our recommendations. Future research should undertake similar examination for the other stages of the research process.

Improving Research Integrity Possibility 2. As the use of AI increases, the need to validate information grows exponentially, resulting in the emerging field of AI self-validation. Both omitting or augmenting data have ethical implications; they introduce bias and make anomalies more difficult to detect. Thus, there will be advantages to using a well-engineered validation process that can provide scalability, consistency, and cost-effective methods of analyzing ever-growing data sets. Adversarial testing uses critical thinking as it finds inaccurate or contradictory information. We believe that ‘data provenance tracking’ might have promising applications for tracking inaccurate citations and tracing inaccurate information across publications, increasing the trust in academic integrity. Data provenance tracking is being discussed in medical, engineering, and technology conferences and journals. But not yet in marketing. Thus, for those unfamiliar with the concept, ‘data provenance tracking’ refers to the process of capturing and recording information about the origin, lineage, and transformations of data throughout its use. Data provenance tracking can help ensure transparency, traceability, and accountability in data-driven processes. By maintaining a record of data sources, modifications, and interactions, organizations and publishers will better understand how data is used, identify potential biases, and address data quality and reliability issues.

Improving Research Integrity Possibility 3. Indicators of AI bias in the Large Language Model have been increasingly hard to detect. Identification of bias is key because as AI adapts, it can also have inaccurate associations, such as all doctors should be male due to underrepresentation in the model. Asymmetric performance is driven by access to a specific culture or geographic location. Sentiment bias can be positive or negative towards certain groups. Language bias can occur for non-standard languages or dialects, which may be interpreted as “incorrect” due to linguistic styles. Historical bias can arise from the roles and events that are seen in current norms. Most harmful are knowledge gaps with limited access to intellectual capital that can be compounded by popularity bias from dominant cultural trends. Combating these biases within the large language models requires systematic testing from diverse inputs to recognize and characterize niche or esoteric information and human intervention in the creation of specialized AI tools to detect bias; it cannot be eliminated completely, but it can be mediated.

Improving Research Integrity Possibility 4. Confirmation bias can be reduced within AI models through data selection techniques, including diverse data collection and analysis to remove expected and biased data outcomes. Rigor in research to question and encourage critical thinking techniques in society effectively reduces bias. Cognitive techniques that question hypotheses and opposite views of the phenomenon studied. Alternatives can be coaxed out through AI analysis and alternate scenarios that would impact the phenomenon in unconventional ways. Cultural cognition needs to be addressed. Core values enhance the diversity of thought and critical thinking, strengthening the validity of the AI model for international use from the use of critical thinking.

Improving Research Integrity Possibility 5. Currently “publishers and editors striving to boost their journal’s impact factors will favor generally more-cited articles with ‘surprising’ findings and disfavor generally less-cited direct replications” (Hyman, 2024, p. 7). More universities are making it mandatory to make data and code publicly available. The Journal of Marketing has adopted this practice. A trend has emerged to register reports of data and codes. It has also been suggested that research teams could replicate certain effects. But what if instead of full-length replication papers, a section of the journal was set aside for replication studies that replicate work that was previously published in that same journal and the papers are limited to a few paragraphs for the print edition with a copy of the dataset, analysis, tables, etc., all appearing as online appendices. That would encourage scholars to engage in replication research for top journals. And have a home for it. That could improve trust in research. Of course, the journals would need to determine how to avoid intentionally failed replication studies by individuals seeking to thwart the research of their rivals (Hyman, 2024). Perhaps a journal that accepts a manuscript with a dataset must agree to publish three replication

studies on that topic. This would permit triangulation of truth, consistent with falsification, scientific realism, and positivism. The academy should serve as a bulwark against assaults on truth as presented by GenAI. The replicating teams cannot have a link to the initial research team. Imagine a group of doctoral students conducting a replication study as part of the professionalization process afforded by doctoral programs. Again, the papers could be limited to a few paragraphs for the print edition with a copy of the dataset, analysis, tables, etc., all appearing as online appendices. That would encourage scholars to engage in replication research for top journals. This possibility at least maintains if not improves trust in research.

Improving Research Collaboration and Writing Quality

Improving Collaboration and Writing Quality Possibility 1. Very recently, individuals had to paste text into GenAI platforms and provide prompts directing the GenAI on how to analyze the text. Now ChatGPT 4.0 can already analyze web links; one pastes the internet hyperlink into the chat and asks their question. It is exciting to envision a future in which the individual scans a QR code or pastes the hyperlink to a webpage and instructs any AI software to analyze the text. Competitive intelligence tools already scrape websites and provide competitive analysis (Spyfu, SimilarWeb, etc.). But when Professor X can simply point the GenAI at the website and instruct it to analyze the descriptions of all the faculty in other departments and colleges at Professor X's university and compare them to the last paper (pasted to the GenAI) that Professor X wrote to identify who would be the best people to collaborate with on a particular (specified) topic, that could help reduce social isolation. Furthermore, academic departments should hold regular meetings to discuss viewpoints on GenAI, how they are using it in classes and research, and what challenges they are facing related to it. One of the authors has been participating in such departmental meetings for the last year held once a semester, and it has proved very beneficial. Through such meetings, best practices can be shared. In so doing, it improves research integrity. Through these meetings, the outcomes help departments navigate the new GenAI-related questions that contain ethical content.

Improving Collaboration and Writing Quality Possibility 2. The impact of GenAI on scholarship for marketing “Bob Rosses.” A young person might be given crayons to draw in elementary school. Eventually, they use colored pencils, instead, which permits better art. Later, they move on to oil pastels and to oil paints. More advanced tools in the hands of master craftspeople are amazing. Can we imagine what Bob Ross could produce using GenAI tools in art? The same applies to marketing and business scholarship. The tools could do amazing things in the hands of distinguished scholars. The effects are much less and much different for individuals who are still operating at a paint-by-numbers level within their research abilities. However, it is often the case that each generation of people struggles with technologies that develop when people are older.

Junior scholars have the opportunity to sit down with senior scholars and together use more advanced prompt engineering, which could be transformative for the marketing field. This should be accomplished with an understanding of the ethical use of the GenAI tool to advance ethical scholarship.

Improving Collaboration and Writing Quality Possibility 3. Usage of GenAI by authors to improve their academic writing. Scholars could use GenAI in a variety of ways to improve their research.¹⁸ They could have it evaluate the spelling, grammar, and reading level. That will improve the research integrity. They could use GenAI as ‘an internal reviewer’—prompting the GenAI to take the role of a reviewer at a particular journal and provide a review of the strengths and weaknesses of the paper. They could use GenAI to provide needed code for analyses. They could use GenAI to serve as a coder for qualitative data categorization. Or they could use two or three different GenAI platforms to code the data and triangulate between with another one. Or they could copy two or three other recently published papers on the same stream of topics and have it compare the paper to the published papers. Many other ideas exist. A growing set of studies has begun to appear at conferences and in journals on the many different ways that GenAI can improve the research process for professors and universities and improve research insights in organizations (e.g., Davenport et al., 2020; Paschen et al., 2019). Successful, trend-setting large firms have begun offering AI assistants to their employees, such as the ‘My Assistant’ to the 50,000 employees at the retail giant Walmart, ‘ChatPwC’ at the leading consulting firm PriceWaterhouseCoopers, and ‘JLL GPT’ to the 103,000 members at the commercial real estate JLL (Wilkinson, 2023). In all of these launches, the main idea is to improve employee decision-making through better insights, not reduce headcount. One question that arises in pursuing this idea is whether it will create even greater differences in resources across different universities for faculty. Or will it equalize them? GenAI can help in idea generation and brainstorming, content creation and drafting, literature searches, the creation of tables and figures, and reviewing a drafted paper to identify key takeaways and contributions. Unless these functions are the personal work of the author and creatively developed, original research may not be the final product. There is always the opportunity to use GenAI functions for unethical purposes, unless there are safeguards to maintain the integrity of the authors’ contributions.

Improving Ethicalness and Quality in Research

Improving Ethicalness and Quality in Research Possibility 1. To decrease the impact in the future of predatory paper mill journals and publishers (which are publishing a large portion of articles containing research integrity problems), perhaps as a part of the review process, the software that reviews papers at journals could (maybe first) review the references. This could include the standard report of whether citations and references match up (that is normally run late in the review process). It

Table 1. How to Create a Better Future in Which Integrity Concerns are Minimized in Scientific Publishing.

GenAI's Recommendation List	Authors' Response to Each of the Recommendations
1. Implement AI-powered detection tools: Develop and widely deploy advanced AI systems capable of identifying manipulated images, fabricated data, and plagiarism.	Yes. A first check that systems should generate for the editor/associate editor to see before even assigning out to reviewers. Similar to a plagiarism check. We believe it would be helpful if it also generated a list indicating the number of 'open access only journals' in the references and the number of listed 'predatory journals' in the references. Scholars have a difficult time on their own as reviewers keeping up to date on the 1000s of journals now in existence as they look at cited materials.
2. Blockchain for data verification: Use blockchain technology to create immutable records of raw data, ensuring transparency and traceability in research.	An interesting idea to explore in more detail. Might this be problematic for data that has to be stored securely to meet HIPAA or other confidentiality requirements?
3. Mandatory data sharing: Require researchers to share their raw data and analysis code as a condition for publication.	As pointed out in the article, GenAI can give you fast, free, 'synthetic' (imaginary) raw data that passes all sniffer tests.
4. Cross-institutional peer review: Establish a system of peer review that involves researchers from multiple institutions and countries to reduce bias and increase scrutiny.	How about instead journals or publishers create and post some very short understandable training videos for reviewers that teach them these things.
5. Research integrity education: Implement comprehensive ethics and integrity training programs for researchers at all career stages, with a focus on cross-cultural understanding.	What if journals and/or publishers have short video training on the topics that all authors on a paper must go through before a paper can be submitted to the journal for review. Don't assume that universities or doctoral programs will consistently do it across programs. Just like most universities have a set of short videos that faculty and staff much watch at the start of each new academic year.
6. Whistleblower protection: Strengthen protections and incentives for individuals who report research misconduct.	What incentives can be provided? Perhaps a scholarship of paying for gold open access for the author's next paper in any one of the publisher's journals?
7. Pre-registration of studies: Expand the use of study pre-registration to reduce the likelihood of data manipulation and p-hacking.	This is not going to work when papers go through several journals over several years before finding a home.
8. Alternative metrics for academic success: Develop and promote new ways of evaluating researchers beyond publication counts, reducing "publish or perish" pressure.	Will that ever work? Let's be real. It is being tried in Spain. But will this create even more of a caste system of journals that keeps scholars from moving up in jobs across institutions?
9. Open science initiatives: Support and expand open science practices, including open access publishing and open peer review, to increase transparency.	Brahahahahaha. Think about this carefully. What is happening right now with open access predatory journals? And how is that influencing the integrity of results in GenAI outputs?
10. International integrity task force: Create a global body to harmonize research integrity standards and facilitate cross-border investigations of misconduct.	That is a good idea. Sort of like the committee that met back in the 1940s to look at the classification of measurement (described by SS Stevens in his famous treatise on the topic of nominal, ordinal, interval, and ratio data).

could also include a summary report on the references. For each paper, first, is that paper retracted or not? Second, is that journal on the predatory journal list? Third, is that journal a 'pay to play' only open access Gold publishing model journal? In addition to the individual level review, a summary report of the number of articles per journal and per the category of predatory journals and 'pay to play' only journals. This report could be generated at the first stage of submission. Perhaps it is made available to authors first, as not all authors on a paper are involved with all aspects and writing of the paper. And not all coauthors know all of the other coauthors really well. Scholars meet new potential coauthors through conferences, and then they work together remotely on a paper, each adding a different value. Sometimes they'll add doctoral students. Retractions due to data fabrication, analysis manipulation, or text plagiarism by one author on the paper affect all the coauthors who might not be aware that one of the contributors to the project either manipulated data or results or plagiarized

text from prior work. Especially for teams from multiple universities. The software results could give insights to the author team so they can fix the situation and provide peer instruction to the collaborator on what was wrong and how they need to change their activities if they want to engage in future work together. Or it could normally be provided to the editor/associate editor and to the reviewers, and the authors might only receive a copy prior to final submission only if they pay a processing fee for it (to prevent bad actors from using the free report to test the system rather than test their article). Many different implementation routes could be taken. For any of them, it would provide valuable information to editors and reviewers. While GenAI and other AI tools have the potential to improve the review process, they should not replace editors or authors. Gendron et al. (2022, p. 1) state, "We are concerned particularly with the gradual removal of human involvement in journal editor and reviewer roles, as artificial intelligence and automated expert systems become increasingly influential across a range

of tasks and judgments historically carried out and performed by people.”

Improving Ethicalness and Quality in Research Possibility 2. As mentioned in the call for papers, new technologies will continue to emerge so rapidly that anticipating all their intended, unintended, and synergistic effects is impossible. At the same time, most would agree with the idea that to “prevent scientific misconduct...researchers should consistently educate themselves regarding ethical principles and standards within their respective academic disciplines” and that “Institutions should provide thorough ethics training programs for both researchers and staff members” (Zhaksylyk et al., 2023, p. 8). In the words of Hyman (2024, p. 15), “We need focus on creating a professional culture of trust.” Departmental conversations on those suggested topics are beneficial, but realistically they probably will not occur across all universities, and those that do occur likely will not be equivalent in topics, viewpoints, and depth. Thus, the training should be accomplished in the future by having a module or set of modules that authors must complete on the publisher’s webpage before authors can submit a paper to any of the journals sponsored by the publisher. And it must be completed individually by every coauthor listed on the paper. Given the great variance in what is considered acceptable behaviors related to honesty across nations, this is really important to establish what is and is not ethical/permissible in the journals. Many junior faculty do not receive any training on that topic in their doctoral programs. And if they do, the topics and views have much variance across institutions around the world. The topic of responsible AI is listed on many companies’ pages of their newly called Chief Artificial Intelligence officers. Scholars deserve training on it as well as industry professionals.

Improving Ethicalness and Quality in Research Possibility 3. Publishers/institutions could band together to provide a training dataset on all of their research articles that scholars could then draw upon. It could be a subscription model for university libraries, similar to the IKON ski pass in which competitor ski resorts allow skiers to ski at all of the mountains across states and countries, depending on the subscription. We are excited when imagining how transformative that could be in changing the rate of scientific advancement. The journal editorial boards would need to agree on how to determine which journals get included in the training dataset. Perhaps something such as the ABDC list and FT50 and other journal ranking lists, with initial weighting based on rankings. While individual publishers are currently creating their unique subscription GenAI tools (such as Scopus AI at Elsevier), what would be more valuable is a tool that looks through the full range of leading journals that are published by multiple publishers.

In ending this section, we note that even more possibilities might exist beyond the 11 listed possibilities. Some might wonder what GenAI would say on the subject, so we asked it. The answers that it provided are found in Table 1 along with our thoughts about its answers in the second column.

Discussion

Sirgy (2025) calls attention in a special issue of the *Journal of Macromarketing* to the importance of marketing addressing the big issues of our time. The *Journal of Marketing* has likewise introduced a set of special issues focused on “Better Marketing for a Better World.” Related to these thrusts, Laczniak and Shultz (2021, p. 201) argue that “If marketers are to create a better world through socially responsible marketing, they require a comprehensive roadmap of what that means and where it leads.” The roadmap provided in this article covers (1) what has been happening in regard to violations of research integrity in scientific research comprehensively across many fields, (2) the potential for GenAI to accelerate that decline in research trustworthiness by malicious, uncaring, and/or uninformed individuals if nothing changes, and (3) alternatives that could be pursued to change that trajectory in the future. It is important to recognize that much of macromarketing research focuses on consumer and societal well-being topics which often draw on findings from research in medical sciences and hard sciences research (Ekici et al., 2021). Thus, research integrity problems and the potential inappropriate usage of AI in publications in those fields impact marketing research. However, it would not be a good roadmap if it didn’t provide navigational guidance. As called out in the H-V model, formal rules of conduct are important to establish by journal editorial boards, by colleges and departments, and by instructors regarding what is ethical usage of GenAI in research. Toward that end, this research article has listed a set of research questions that might help advance needed discussions in the contexts of journal editorial board discussions, college and department administrative meetings, and faculty instruction to students. While the GenAI technology itself will change over time, the questions and normative guidelines outlined in this article will still apply.

In addition to problems that can occur within generated text, GenAI can also generate data that can match nearly any desired relationship. The recent journal requirements to publish datasets to enable third party result replication are not sufficient to ensure research integrity. Alternatives are presented to help the field change the trajectory of the rising research integrity problems. Pandora’s box has been opened. This means that GenAI can and is being used for unethical purposes. Attempting to close the box would be impossible due to the ongoing race to further develop GenAI’s abilities in augmenting scientific research.

Many journals and COPE are now instructing authors that ChatGPT and other GenAI tools cannot be listed as coauthors on research submissions and have provided very basic policies on GenAI’s ethical use. Perhaps we would collectively alter our course toward a better future if we started thinking of GenAI software tools as an instrument to improve critical thinking and the creative development of research outcomes. In conjunction with the list of outlined possibilities in the preceding section, it would be helpful if marketing journals, societies, and leaders, such as the directors of doctoral programs, provided deeper education and training on the ethical use of GenAI.

Other solutions are, of course, needed to deal with uncaring Machiavellian researchers seeking to maximize research grants and fame, or with those trying to subvert science in cognitive warfare attempts, etc. As we think about the future, we want to pursue regarding research integrity and responsible GenAI usage, we believe it is worth revisiting the question posed by Holbrook and Hulbert (2002): What *do* we produce in the knowledge factory and for whom?

There needs to be improvement in the existing research culture of publishing how academics learn how to get published. There has been an overstatement on what current GenAI is capable of doing; Saba (2024, p. 1) shows several case examples that demonstrate that “language understanding capabilities have been widely exaggerated. While LLMs have proven to generate human-like coherent language (since that’s how they were designed), language understanding capabilities” have a lot of improvement to make. If LLMs that power GenAI do not understand what they are doing, but they are producing human-like coherent language, are they a useful mirror for ethical self-examination of the different personal characteristics? In reflecting on that usefulness, assume the debated question on whether a set of monkeys on typewriters might at some point type a Shakespeare play. If they did, would we value the output just as much regardless of whether the monkeys, Shakespeare, or someone else was the author? And, as pointed out by Ferrell et al. (2024), it is very important to have a theoretical framework like an adaptation of the H-V model to guide AI ethics in marketing research.

A decade ago, every media outlet, company, and business school was clamoring about ‘big data’. Then came blockchain. Indeed, since the beginning of what constitutes ‘modern marketing’ history more than a century ago, marketing has been driven by technological innovations (Hunt, 2018; Hunt et al., 2021). Do we believe that GenAI is more transformative than most of those prior technologies? It might be tempting to be dismissive. IBM president Tom Watson once stated, “I think there is a world market for maybe five computers.” Digital equipment president Ken Olsen once stated, “There is no reason for any individual to have a computer in their home.” Microsoft president Bill Gates said in 1994, “I see little commercial potential for the internet for the next 10 years.” Their forecasts were a little off. Will there be 10 people in the world who *are not* using GenAI at some point in the future? Already an estimated two-thirds of adults have used GenAI to answer questions (Bianchi, 2024; Thormundsson, 2023). Perhaps the current hype surrounding GenAI technologies will experience a ‘bubble burst’ similar to what happened in regards to e-commerce technologies with the dot.com bubble burst twenty five years ago; but e-commerce continues to be an important part of our world (Hagberg et al., 2020; Watson & Wu, 2022). And we believe that GenAI will also continue to be an important part of our world after the hype fades away.

Technology is always evolving. People once protested moving away from using slide rules to calculators and from using typewriters to computers. The real question for the future isn’t about the technology itself—it is about who we are in

our hearts and choices as researchers. What percentage will choose to seek to use the technology ethically when “no one is looking”? What we need is ethical reform across all nations. Macromarketing research is needed on how to get there: what are the moral foundations of society? And when someone in the future types that question into a GenAI platform to get help in answering the question, can we critically evaluate the response, especially if the GenAI answers ‘42’?¹⁹

Conclusions

GenAI has been embraced by the academic community with nearly all marketing conferences and many special issues at journals promoting it as a part of their themes. A growing stream of recent macromarketing research has begun to discuss AI and GenAI (e.g., Dholakia et al., 2025; Grewal et al., 2024a; Hyman & Lukosius, 2024; Rindfleisch et al., 2024; Ferrell & Ferrell, 2024a; Shankar, 2024). Marketing scholarship has looked at many themes in marketing ethics over the last several decades (Schlegelmilch & Öberseder, 2010). We argue that GenAI is one of the most important new themes in marketing ethics, and thus research on ethics and GenAI needs to be discussed in the most prominent marketing journals. Moreover, research on GenAI can help fulfill macromarketing research’s mission of saving the world “amidst significant changes in the provisioning technology which was transforming markets and marketing” (Ekici et al., 2021, p. 25) given that GenAI meets the criteria of being a technology that is transforming markets and marketing. This might be achieved through a cross-disciplinary focus on societal development (Peterson, 2006), “thinking broadly across subject matters” (Ekici et al., 2021, p.27), and “realizing a desirable future” (Hyman & Kostyk, 2022, p. 1).

Indeed, while the uninformed, lazy, or intentionally deceptive use of GenAI poses several risks to the integrity of academic research, the appropriate use of GenAI offers numerous opportunities to enhance the academic research process and accelerate both discovery and justification (verification). While we must be aware that GenAI has the potential to be used as a tool to facilitate misconduct, including data fabrication, falsification, plagiarism, and support for predatory journals, we must also focus more on GenAI’s potential to contribute to a better world by supporting the enhanced ethical development of knowledge. In choosing between these two potentials, scholars face challenges from the publishing culture “publish or perish” to obtain tenure or financial rewards. This environment exists within most doctoral programs providing little or no guidance on publishing ethics. That needs to change. The AMA, AMS, and most other journals have publishing policies and codes of ethics. But Enron also had a detailed, published code of ethics (Ferrell, 2016). As highlighted in the H-V model of ethical decision making, people draw on a combination of the informal, unstated actual practices combined with the published formal codes in evaluating the guidelines from their department, university, area of research community, and journals in determining the deontological norms of right

and wrong. Better communication of policies, codes, and the need for required education, enforcement, and compliance remains necessary, but not sufficient. Published dialogue is needed that discusses the many different ways in which GenAI might be ethically used during the research process (such as the bullet point list of questions found earlier in this article) so that newer scholars, as well as scholars newer to GenAI, can have better normative guidance. Currently, doctoral students and professors most often learn from others 'best practices' on how to get published. This environment can result in a naive decision or intentionally engaging in misconduct. GenAI will continue to improve its abilities, and it will become more difficult to detect in submissions. Faculty are often not able to identify academic research that is the product of GenAI. The fabrication of data sets and experiments poses a threat as technology advances. Possibilities for approaching GenAI tools with care and responsibility are presented in this paper to maintain accountability and minimize the risks of misuse. The marketing discipline should play a leadership role in making GenAI a positive force for advancing knowledge discovery and verification.

Near the completion of an invited revision of this manuscript, we pondered the question: what might some of the original great thinkers of macromarketing who wrote on ethics and marketing prior to their departure from scholarly writing say about the usage of GenAI in academic writing? We posed the question to GenAI, asking it to outline what Shelby D. Hunt, a noted legend in marketing (see, e.g., Ferrell & Ferrell, 2024b; Hansen, 2024) and one of the several founders of the *Journal of Macromarketing*, might say on it. We, the authors, then pondered the likely accuracy of the GenAI's output; we note that four of us in the author team had personal interactions with Shelby over many years. To summarize, the GenAI output described Shelby's commitment to scientific realism's emphasis that successful scientific theories increasingly approximate objective truth about observable and unobservable phenomena. The output indicated that Shelby might view GenAI as a valuable tool for hypothesis generation and pattern identification. We agree that is consistent with Tenet 1 of scientific realism that "the world exists independent of human knowledge of it" (e.g., Hansen, 2024, p. 1197). However, the GenAI output also said Shelby might be concerned about researchers adopting AI-generated content as authoritative without engaging in critical evaluation, which we note is consistent with Tenet 3 that "All knowledge claims must be critically evaluated and tested to determine to what extent they do or do not represent or correspond to the world" (e.g., Hansen, 2024, p. 1197). We call out that there is a difference between having GenAI make a critical evaluation and the authors themselves engaging in a critical evaluation of the GenAI critical evaluation—this can become a 'chicken-and-egg' iterative spiral. How does one keep it from becoming an echo chamber or introducing novel, good-sounding excrement into the conversation? (See, e.g., Hicks et al., 2024). Furthermore, according to the output, Shelby would likely argue that GenAI should be used for knowledge advancement rather than replacing the fundamental research

process. The output indicated that Shelby's emphasis on objectivity (see, e.g., Hunt, 1993; Hunt, 2016) would lead Shelby to raise concerns regarding potential biases that might be embedded in training data of some GenAI systems. The output suggested that Shelby's views might result in him asking whether GenAI outputs contain genuine approximations of truth or if they just contain statistically probable text patterns.

Last, the output suggested that Shelby's pragmatic emphasis might lead him to appreciate GenAI's potential for helping scholars when it comes to the 'context of discovery' related activities (see, e.g., Hunt, 1993; Hunt, 2016), including literature reviews, identifying research gaps, and suggesting methodological approaches - provided researchers engage in critical evaluation (tenet 2). Shelby would likely insist that researchers clearly delineate between AI-assisted work and their own original contributions, maintaining the integrity of scientific attribution that underpins his vision of progressive scientific knowledge.

But herein lies a crux: when GenAI becomes increasingly integrated across several steps of the research process—and more deeply embedded into the most basic tools (Word, Excel, etc.) at a nearly indistinguishable level, how can scholars clearly list it—and do we want a full page or two of description of all GenAI usage in the research. When (and we are *not* suggesting right now is the time) do we move beyond the flagging of it—to a focus back on the 'basics' related to integrity related to trust, equity, and responsibility in how to actually wisely use these new tools (see, e.g., Abela & Murphy, 2008; Gundlach & Murphy, 1993; Moorman et al., 1993; Morgan & Hunt, 1994; Murphy et al., 2007; Nill & Schibrowsky, 2007). In the adoption of the good practices outlined in this article, one of the big questions is will doing so open up more time in their schedules for faculty to engage in additional face-to-face mentoring of students and collaboration with administration, industry practitioners, and policy makers? And/or will there be a GenAI race in quantity of submissions with increased quantity of publication standards for tenure and promotion and/or pressures associated with reductions in workforce in universities?

We believe that when used with the right mindset and awareness, GenAI can be transformative in our research contributions. As we worked on this article, there were several instances in which we took ideas that we had drafted and had GenAI provide critique and evaluation of our ideas. We used dozens of different GenAI prompts in combination with traditional search in the literature review process, both in the initial review and at each new iteration in writing and rewriting to search for potentially relevant references; we located and read the original source materials that the GenAI referenced. We utilized GenAI to examine the reading level, flow, effective rhetoric, tone, voice, transitions, spelling, grammar, and other writing elements. We ran each of those as separate prompts and asked it to repeat the review doing a better job, telling it the analysis was really important. And then we made improvements on some of the responses that it indicated. This did not replace the usage of internal reviewers, which we used, or the usage of professional copy editors, which we also used. GenAI can be used as a tool to iteratively examine the

refinements to the same text paragraph dozens of times—we would not ever ask an internal reviewer or professional copy editor to do that same thing. The usage of GenAI in this article did speed up the research process—but it did not shorten the time of the research process—because we used that same length of time to go much deeper into the ‘weeds’ of different activities than we could a few years ago.

And while industry practitioners should rightfully be concerned from a trade secret, confidentiality, competitive advantage, etc., standpoint about usage of open GenAI platforms that might take their prompt content (even questions) and recycle that back into the training data that is then used in interactions with other people (e.g., competitors) around the world—that dissemination is the whole point of what we *as scholars* are trying to do—to influence thinking of other people, advancing science and society. And if our research *hopefully* is published at some point, others who used GenAI that gave them answers to prompts that drew on our ideas that we had previously included in the prompts as we constructed the paper, they will eventually find and cite our work. This however brings up a concern—if most ‘good’ research in reputable journals in the future is firewall protected from GenAI inclusion in its training data while other research in predatory and questionable journals is available, both to GenAI for its training data, as well as to consumers as their new search tool (skipping Google and Wikipedia altogether), what does this mean for knowledge and decision-making among scholars, policy makers, managers, and consumers who do not have access to the firewall protected research findings? Will GenAI help elevate the knowledge of those globally who are facing tougher socioeconomic challenges? Or will it increase the divide in knowledge as it relates to what is true?

Associate Editor

Michael Hyman

ORCID iDs

Jared M Hansen  <https://orcid.org/0000-0003-1593-5418>

Linda Ferrell  <https://orcid.org/0000-0002-2309-309X>

Michael A Levin  <https://orcid.org/0000-0002-0046-1988>

Paul Wilson  <https://orcid.org/0009-0007-9838-9470>

Kellie Cloud  <https://orcid.org/0009-0002-8396-6523>

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Statement on LLM and Gen AI

Gen AI was used in the creation of example content presented in Figures 1 and 2 and Table 1. Gen AI was used for copy editing review and to identify additional relevant literature, as described in the article.

Notes

1. We note awareness of the dispute over the origin of this quote; the dispute itself, in addition to the quote, is also consistent with the paper's thrust on the role that GenAI might play in the future on the increasing problems in research integrity.
2. Recent counts indicate China has approximately 2,700 higher education institutions (according to the China Ministry of Education, 2020), Russia has about 700 universities (according to Statista online), and India has over 1,000 universities (according to the India University Grants Commission online). For reference, the United States has approximately 2,600 four-year higher education institutions (according to the NCES, 2020).
3. the situation received viral global coverage when media was published showing parents climbing the outside of a building to provide cheat sheets to students taking exams (Lakshmi 2015).
4. The National Science Boards reports the following science and engineering-related statistics in their Table PBS-1: China (2012) = 329,067, United States (2012) = 430,164, China (2022) = 898,949, United States (2022) = 457,335. <https://nces.nsf.gov/pubs/nsb202333/publication-output-by-region-country-or-economy-and-by-scientific-field#output-by-region-country-or-economy>
5. We acknowledge a multitude of factors beyond research integrity (such as perhaps recent campus protests and political agendas, tuition increases, accusations of institutional adoption of wokeism and moral relativism leading lack of free speech on campus, failure to teach needed, relevant skills, etc.) that likely contribute to this trust erosion. However, these issues fall outside the scope of this research focus. See, e.g., Nietzel (2024).
6. We thank Dr. Stewart for his consent to be quoted in this research.
7. We agree with the idea put forth by Kotler and Levy (1969, p. 10) that ‘marketing is a pervasive societal activity that goes considerably beyond the selling of toothpaste, soap, and steel.’ Reflecting on that article, Kotler (2024) states “We said marketing activities occur everywhere in life.” Moreover, marketing is perhaps the best suited discipline to examine a wide range of social issues to “realize a desirable future” (Hyman & Kostyk, 2022, p. 1).
8. https://scholar.google.com/scholar?hl=en&as_sdt=5%2C45&sciodt=0%2C45&cites=2431687974108665522&scipsc=&as_ylo=2012&as_yhi=2022
9. https://scholar.google.com/scholar?hl=en&as_sdt=5%2C45&sciodt=0%2C45&cites=2431687974108665522&scipsc=&as_ylo=2023&as_yhi=
10. <https://us.sagepub.com/en-us/nam/artificial-intelligence-policy>
11. <https://www.elsevier.com/about/policies-and-standards/the-use-of-generative-ai-and-ai-assisted-technologies-in-writing-for-elsevier>
12. <https://taylorandfrancis.com/our-policies/ai-policy/>
13. <https://www.wiley.com/en-us/publish/book/ai-guidelines>
14. <https://academic.oup.com/pages/for-authors/books/author-use-of-artificial-intelligence>
15. <https://www.cambridge.org/core/journals/flow/information/journal-policies/publishing-ethics>
16. <https://www.springer.com/gp/editorial-policies/artificial-intelligence-ai-/25428500>
17. <https://labs.google.com/search?source=hp&authuser=0> accessed on July 27, 2024

18. Another example is how we used GenAI to assist us as we wrote this section. Originally, we had six concerns listed in the section on future concerns and 11 opportunities in the section on future opportunities. A reviewer encouraged us to find a way to have more parallel construction between the two major sections. So we copied what we had written into GenAI with the Prompt 1: “Assume you are a scholar revising the following copied text. It has two sections. The first section is on future challenges. The second section is on future possibilities. The instructions from journal reviewers are that the challenges and possibilities sections should be balanced in terms of how many are in both sections. And they should somewhat relate to each other. With those instructions, and in an attempt to keep all the text and ideas in the present draft, create an outline of the new subheadings in both sections. It is okay to move material between the two sections where appropriate.” Then we gave it prompt 2: “Adopting the outline headers that you have provided, include all of the original text into that new outline under the best matching headings. Don’t leave anything out. This is really important.” The software did as requested. We copied the output to a new Word document. Then we looked at the original paper and the suggested reorganization side by side (two different documents) on the computer monitor. We then moved around things in the *original* document in a way that we liked, taking into account the *suggestions* of the GenAI tool output.
19. In regards to this closing question on what are the moral foundations of society, if GenAI replied with ‘42’—the answer provided in Adam’s (1980) novel by the ‘Deep Thought’ supercomputer after 7.5 million years to the question what is “the Ultimate Question of Life, the Universe, and Everything”—how would we interpret it? As mentioned on some forums, the number 42 is ASCII code for the star symbol (*) which is a ‘wildcard’ denoting anything or everything; in regards to societal moral foundations, is that insightful or AI hallucination? Is it a good moral principle or is it nihilistic?

References

- Abela, A. V., & Murphy, P. E. (2008). Marketing with integrity: Ethics and the service-dominant logic for marketing. *Journal of the Academy of Marketing Science*, 36(1), 39–53. <https://doi.org/10.1007/s11747-007-0062-0>
- Adams, D. (1980). *The hitchhiker’s guide to the galaxy*. Harmony Books.
- Akaka, M. A., Vargo, S. L., Nariswari, A., & O’Brien, M. (2023). Microfoundations for macromarketing: A metatheoretical lens for bridging the micro-macro divide. *Journal of Macromarketing*, 43(1), 61–75. <https://doi.org/10.1177/02761467211054349>
- Ameen, N., Sharma, G. D., Tarba, S., Rao, A., & Chopra, R. (2022). Toward advancing theory on creativity in marketing and artificial intelligence. *Psychology and Marketing*, 39(9), 1802–1825. <https://doi.org/10.1002/mar.21699>
- Bailey, J. (2024, January 8). The weaponization of plagiarism. Plagiarism Today. <https://www.plagiarismtoday.com/2024/01/08/the-weaponization-of-plagiarism-2/>
- Baker, T. (2023, July 19). Stanford president resigns over manipulated research, will retract at least three papers. Stanford Daily. <https://stanforddaily.com/2023/07/19/stanford-president-resigns-over-manipulated-research-will-retract-at-least-3-papers/>
- Baughman, J. (2023, October 21). China’s ChatGPT war. <https://www.airuniversity.af.edu/Portals/10/CASI/documents/Research/Cyber/2023-08-21%20China’s%20ChatGPT%20War.pdf>
- Bech-Larsen, T., & Aschemann-Witzel, J. (2012). A macromarketing perspective on food safety regulation: The Danish ban on trans-fatty acids. *Journal of Macromarketing*, 32(2), 208–219. <https://doi.org/10.1177/0276146711435262>
- Bianchi, T. (2024, September 10). U.S. adults using generative AI usage first for online search 2023-2027. <https://www.statista.com/statistics/1454204/united-states-generative-ai-primary-usage-online-search/>
- Bik, E. M., Casadevall, A., & Fang, F. C. (2016). The prevalence of inappropriate image duplication in biomedical research publications. *mBio*, 7(3), e00809–16. <https://doi.org/10.1128/mbio.00809-16>
- Blanding, M. (2019, February 6). The strange case of Dr. Ho Man Kwok. Colgate Magazine. <https://news.colgate.edu/magazine/2019/02/06/the-strange-case-of-dr-ho-man-kwok/>
- Borsboom, D., & Wagenmakers, E. J. (2012). Derailed: The rise and fall of Diederik Stapel. *APS Observer*, 26. <https://www.psychologicalscience.org/observer/derailed-the-rise-and-fall-of-diederik-stapel>
- Boychev, H. (2021, April 8). Top German psychologist fabricated data, investigation finds. *Science (New York, N Y)*. <https://www.science.org/content/article/top-german-psychologist-fabricated-data-investigation-finds>
- Bozkurt, A., Junhong, X., Lambert, S., Pazurek, A., Crompton, H., Koseoglu, S., Romero-Hall, E., Bond, M., Nerantzi, C., Honeychurch, S., Bali, M., Dron, J., Mir, K., Stewart, B., Costello, E., Mason, J., Stracke, C. M., Romero-Hall, E., Koutropoulos, A., . . . Jandrić, P. (2023). Speculative futures on ChatGPT and generative artificial intelligence (AI): A collective reflection from the educational landscape. *Asian Journal of Distance Education*, 18(1), 53–130. <https://asianjde.com/ojs/index.php/AsianJDE/article/view/709>
- Brainard, J. (2023). ‘Hijacked’ journals mar top database. *Science (New York, N Y)*, 382(6675), 1102. <https://doi.org/10.1126/science.adn3529>
- Bratton, V. K., & Strittmatter, C. (2013). To cheat or not to cheat?: The role of personality in academic and business ethics. *Ethics & Behavior*, 23(6), 427–444. <https://doi.org/10.1080/10508422.2013.811077>
- Breakwell, G. M. (2020). Mistrust, uncertainty and health risks. *Contemporary Social Science*, 15(5), 504–516. <https://doi.org/10.1080/21582041.2020.1804070>
- Breckler, S. J. (2012). Faking it. *Monitor on Psychology*, 43(8), 56. <https://www.apa.org/monitor/2012/09/sd>
- Brush, G. J., Guo, X., Hunting, A., & Frethey-Bentham, C. (2024). Using qualitative comparative analysis to identify complex solutions and optimal combinations of conditions influencing COVID vaccine acceptance: A primer for QCA. *Journal of Macromarketing*, 44(2), 276–306. <https://doi.org/10.1177/02761467231182300>
- Buntz, B. (2023, May 2). 10 pioneering companies implementing AI in drug discovery, development and beyond. <https://www.>

- drugdiscoverytrends.com/10-pioneering-companies-implementing-ai-in-drug-discovery-development-and-beyond/
- Cabanac, G., Clause, A., Jégou, L., & Maisonnobe, M. (2023, September). The geography of retracted papers: Showcasing a CrossRef–dimensions–netcity pipeline for the spatial analysis of bibliographic data. STI 2023: 27th international conference on science, technology and innovation indicators.
- Callaway, E. (2011). Report finds massive fraud at Dutch universities. *Nature*, 479(7371), 15. <https://doi.org/10.1038/479015a>
- Cao, C., Li, N., Li, X., & Liu, L. (2013). Reforming China's S&T system. *Science (New York, N Y)*, 341(6145), 460–462. <https://doi.org/10.1126/science.1234206>
- Casal, J. E., & Kessler, M. (2023). Can linguists distinguish between ChatGPT/AI and human writing?: A study of research ethics and academic publishing. *Research Methods in Applied Linguistics*, 2(3), 100068. <https://doi.org/10.1016/j.rmal.2023.100068>
- Cengage. (2024, October 2). Faces of faculty: The higher education instructor experience 2024. *Cengage*. <https://blog.cengage.com/2024-faces-of-faculty-the-higher-education-instructor-experience/>
- Charlebois, S., & Labrecque, J. (2009). Sociopolitical foundations of food safety regulation and the governance of global agrifood systems. *Journal of Macromarketing*, 29(4), 363–373. <https://doi.org/10.1177/0276146709346255>
- ChatGPTGuide. (2024, July 16). Apple, Nvidia under fire for using YouTube videos to train AI without consent. *ChatGPTGuide.com*. <https://www.chatgptguide.ai/2024/07/16/apple-nvidia-under-fire-for-using-youtube-videos-to-train-ai-without-consent/>
- China Ministry of Education. (2020, October 6). Overview of educational achievements in China in 2019. http://en.moe.gov.cn/documents/reports/202102/t20210209_513095.html
- Chowdhury, R. M. (2020). Personal values and consumers' ethical beliefs: The mediating roles of moral identity and machiavellianism. *Journal of Macromarketing*, 40(3), 415–431. <https://doi.org/10.1177/0276146720917431>
- Chowdhury, R. M. (2023). Perceived income inequality, trust, and consumers' ethical judgments. *Journal of Macromarketing*, 43(4), 476–493. <https://doi.org/10.1177/02761467231187306>
- Christensen, J., Hansen, J. M., & Wilson, P. (2024). Understanding the role and impact of generative artificial intelligence (AI) hallucination within consumers' tourism decision-making processes. *Current Issues in Tourism*, 28(4), 1–16. <https://doi.org/10.1080/13683500.2023.2300032>
- Chui, M., Manyika, J., Miremadi, M., Henke, N., Chung, R., Nel, P., & Malhotra, S. (2018, April 17). Notes from the AI frontier: Applications and value of deep learning. McKinsey. <https://www.mckinsey.com/featured-insights/artificial-intelligence/notes-from-the-ai-frontier-applications-and-value-of-deep-learning>
- Claverie, B., & Du Cluzel, F. (2022). "Cognitive warfare": The advent of the concept of "cognitics" in the field of warfare. In B. Claverie, B. Prébot, N. Buchler, & F. du Cluze (Eds.), *Cognitive warfare: The future of cognitive dominance* (vol. 2, pp. 1–10). NATO Collaboration Support Office. <https://hal.science/hal-03635889>
- Crawford, J., Cowling, M., & Allen, K. A. (2023). Leadership is needed for ethical ChatGPT: Character, assessment, and learning using artificial intelligence (AI). *Journal of University Teaching & Learning Practice*, 20(3). <https://doi.org/10.53761/1.20.3.02>
- Cressey, D. R. (1953). *Other people's money; a study of the social psychology of embezzlement*.
- Crittenden, V. L., Hanna, R. C., & Peterson, R. A. (2009). The cheating culture: A global societal phenomenon. *Business Horizons*, 52(4), 337–346. <https://doi.org/10.1016/j.bushor.2009.02.004>
- Crocker, J., & Cooper, M. L. (2011). Addressing scientific fraud. *Science (New York, N Y)*, 334(6060), 1182–1182. <https://doi.org/10.1126/science.1216775>
- Cyranoski, D. (2018, December 14) China introduces 'social' punishments for scientific misconduct. *Nature*. <https://doi.org/10.1038/d41586-018-07740-z>
- Dahlberg, B. (2018, September 26). Cornell food researcher's downfall raises larger questions for science. *NPR*. <https://www.npr.org/sections/thesalt/2018/09/26/651849441/cornell-food-researchers-downfall-raises-larger-questions-for-science>
- Dalalah, D., & Dalalah, O. M. A. (2023). The false positives and false negatives of generative AI detection tools in education and academic research: The case of ChatGPT. *International Journal of Management Education*, 21(2), 10822. <https://doi.org/10.1016/j.ijme.2023.1>
- D'Alessio, F. A. (2021). Computational propaganda: Challenges and responses. *Academia Letters*, 10, 1–8. <https://doi.org/10.20935/AL3468>
- Davenport, T., Guha, A., Grewal, D., & Bressgott, T. (2020). How artificial intelligence will change the future of marketing. *Journal of the Academy of Marketing Science*, 48(1), 24–42. <https://doi.org/10.1007/s11747-019-00696-0>
- Davis, M. S. (1971). That's interesting! towards a phenomenology of sociology and a sociology of phenomenology. *Philosophy of the Social Sciences*, 1(2), 309–344. <https://doi.org/10.1177/004839317100100211>
- Day, G. S., & Malcolm, R. (2012). The CMO and the future of marketing. *Marketing Management*, 22(Spring), 34–43.
- Deer, B. (2020). *The doctor who fooled the world: Science, deception, and the war on vaccines*. Johns Hopkins University Press.
- Devereaux, Z. P., Lecocq, R., Labrie, M. A., & Forrester, B. C. (2023). Contemporary media and the AI-goldrush from a military perspective. Annual IDEaS conference: Disinformation, hate speech, and extremism online. <https://www.cmu.edu/ideas-social-cybersecurity/events/ideas-ai-goldrush-and-military-perspective-v1.6.pdf>
- Dholakia, N., Repo, P., Firat, A. F., Campos, I., & Timonen, P. (2025). Artificial intelligence, structure of knowledge, and the future directions for macromarketing. *Journal of Macromarketing*, 45(2), 337–344. <https://doi.org/10.1177/02761467251323840>
- Domegan, C., Brychkov, D., McHugh, P., McNamara, Á, Harkin, K., Fitzgerald, C., & O'Donovan, D. (2020). Marketing systems: A listen, learn, leverage framework. *Journal of Macromarketing*, 40(3), 380–395. <https://doi.org/10.1177/0276146720922282>
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., & Lim, W. M. (2021). How to conduct a bibliometric analysis: An overview and guidelines. *Journal of Business Research*, 133, 285–296. <https://doi.org/10.1016/j.jbusres.2021.04.070>
- Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28). <https://doi.org/10.1126/sciadv.adn5290>

- Doss, C., Mondschein, J., Shu, D., Wolfson, T., Kopecky, D., Fitton-Kane, V. A., Bush, L., & Tucker, C. (2023). Deepfakes and scientific knowledge dissemination. *Scientific Reports*, 13(1), 13429. <https://doi.org/10.1038/s41598-023-39944-3>
- Eddy, N. (2024, December 20). What is data poisoning, and how can you prevent it? *BizTech*. <https://biztechmagazine.com/article/2024/12/what-is-data-poisoning-perfcon>
- Ekici, A., Genc, T. O., & Celik, H. (2021). The future of macromarketing: Recommendations based on a content analysis of the past twelve years of the journal of macromarketing. *Journal of Macromarketing*, 41(1), 25–47. <https://doi.org/10.1177/0276146720966654>
- Elkhatat, A. M., Elsaid, K., & Almeer, S. (2023). Evaluating the efficacy of AI content detection tools in differentiating between human and AI-generated text. *International Journal for Educational Integrity*, 19(17). <https://doi.org/10.1007/s40979-023-00140-5>
- Fang, F. C., Steen, R. G., & Casadevall, A. (2012). Misconduct accounts for the majority of retracted scientific publications. *Proceedings of the National Academy of Sciences*, 109(42), 17028–17033. <https://doi.org/10.1073/pnas.1212247109>
- Fass-Holmes, B. (2018). International students reported for academic integrity violations: Demographics, retention, and graduation. *Journal of International Students*, 7(3), 644–669. <https://doi.org/10.32674/jis.v7i3.292>
- Felin, T., & Holweg, M. (2024). Theory is all you need: AI, human cognition, and decision making. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4737265
- Ferrell, L. (2021). Big thinking about marketing. *Journal of Macromarketing*, 41(4), 527–530. <https://doi.org/10.1177/02761467211053533>
- Ferrell, L., & Ferrell, O. C. (2022). Broadening the definition of socially responsible marketing. *Journal of Macromarketing*, 42(4), 560–566. <https://doi.org/10.1177/02761467221094570>
- Ferrell, O. C. (2016). A framework for understanding organizational ethics. In *Business ethics* (pp. 3–17). Routledge.
- Ferrell, O. C., & Ferrell, L. (2008). A macromarketing ethics framework: Stakeholder orientation and distributive justice. *Journal of Macromarketing*, 28(1), 24–32. <https://doi.org/10.1177/0276146707311290>
- Ferrell, O. C., & Ferrell, L. (2024a). Building a better world: The role of AI ethics and social responsibility. *Journal of Macromarketing*, 44(4), 928–935. <https://doi.org/10.1177/02761467241285793>
- Ferrell, O. C., & Ferrell, L. (2024b). Trailblazing the path for marketing ethics: The profound influence of Shelby Hunt. *Journal of Marketing Management*, 40(13–14), 1174–1192. <https://doi.org/10.1080/0267257X.2023.2295273>
- Ferrell, O. C., Fraedrich, J., & Ferrell, L. (2022). *Business ethics: Ethical decision making and cases*. Cengage.
- Ferrell, O. C., & Gresham, L. G. (1985). A contingency framework for understanding ethical decision making in marketing. *Journal of Marketing*, 49(3), 87–96. <https://doi.org/10.2307/1251618>
- Ferrell, O. C., Gresham, L. G., & Fraedrich, J. (1989). A synthesis of ethical decision models for marketing. *Journal of Macromarketing*, 9(2), 55–64. <https://doi.org/10.1177/027614678900900207>
- Ferrell, O. C., Harrison, D. E., Ferrell, L. K., Ajjan, H., & Hochstein, B. W. (2024). A theoretical framework to guide AI ethical decision making. *AMS Review*, 14(3), 1–15. <https://doi.org/10.1007/s13162-024-00275-9>
- Finkel, M. L., & Rosen, T. (2007). Disease prevention through vaccination: The science and the controversy. In M. L. Finkel (Ed.), *Truth, lies, and public health: How we are affected when science and politics collide* (pp. 371–394). Praeger.
- Gelman, A. (2024, February 6). It's bezzle time: The dean of engineering at the University of Nevada gets paid \$372,127 a year and wrote a paper that's so bad, you can't believe it. <https://statmodeling.stat.columbia.edu/2024/02/06/its-bezzle-time-the-dean-of-engineering-at-the-university-of-nevada-gets-paid-372127-a-year-and-wrote-a-paper-thats-so-bad-you-cant-believe-it-i-mean-really-you-have-to-take-a-look-at-t/>
- Gendron, Y., Andrew, J., & Cooper, C. (2022). The perils of artificial intelligence in academic publishing. *Critical Perspectives on Accounting*, 87, 102411. <https://doi.org/10.1016/j.cpa.2021.102411>
- Gonsalves, A. (2024, August 29). Who is Debarghya Das, the Indian-origin influencer being slammed over his guide to get US citizenship? *Masala!* <https://www.masala.com/creators-corner/who-is-debarghya-das-the-indian-origin-influencer-being-slammed-over-his-guide-to-get-us-citizenship>
- Grewal, D., Guha, A., & Becker, M. (2024a). AI is changing the world: For better or for worse? *Journal of Macromarketing*, 44(4), 870–882. <https://doi.org/10.1177/02761467241254>
- Grewal, D., Satomino, C. B., Davenport, T., & Guha, A. (2024b). How generative AI is shaping the future of marketing. *Journal of the Academy of Marketing Science*, 53, 1–21. <https://doi.org/10.1007/s11747-024-01064-3>
- Grewal, R. (2017). Journal of marketing research: Looking forward. *Journal of Marketing Research*, 54(1), 1–4. <https://doi.org/10.1509/jmr.54.11>
- Grigorieva, N. M., & Petrenko, S. A. (2023, November). Known adversarial attacks by 2023. In *2023 Seminar on information systems theory and practice (ISTP)* (pp. 32–36). IEEE.
- Grimes, P. W. (2004). Dishonesty in academics and business: A cross-cultural evaluation of student attitudes. *Journal of Business Ethics*, 49(February), 273–290. <https://doi.org/10.1023/B:BUSI.0000017969.29461.30>
- Guan, M. (2016, March 2). Comment on: Yang, Wei (2016). Research integrity in China. *Science (New York, N Y)*. <https://www.science.org/doi/full/10.1126/science.1247700>
- Gundlach, G. T., & Murphy, P. E. (1993). Ethical and legal foundations of relational marketing exchanges. *Journal of Marketing*, 57(4), 35–46. <https://doi.org/10.1177/00222429930570040>
- Hagberg, J., Kjellberg, H., & Cochoy, F. (2020). The role of market devices for price and loyalty strategies in 20th century US grocery stores. *Journal of Macromarketing*, 40(2), 201–220. <https://doi.org/10.1177/0276146719897366>
- Hansen, J. M. (2024). The philosophical foundations of ethical decision-making: Past controversies, current misunderstandings, and future opportunities. *Journal of Marketing Management*, 40(13–14), 1193–1223. <https://doi.org/10.1080/0267257X.2024.2421850>
- Harari, Y. N. (2016). *Homo deus: A brief history of tomorrow*. Random House.

- Herndon, N. (2016). Research fraud and the publish or perish world of academia. *Journal of Marketing Channels*, 23(3), 91–96. <https://doi.org/10.1080/1046669X.2016.1186469>
- Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, 26(38), 1–10. <https://doi.org/10.1007/s10676-024-09775-5>
- Hill, T. P. (1999). The difficulty of faking data. *Chance*, 12(3), 27–31. <https://doi.org/10.1080/09332480.1999.10542154>
- Holbrook, M. B., & Hulbert, J. M. (2002). Review: What do we produce in the 'knowledge factory' and for whom? A review essay of 'the knowledge factory' by Stanley Aronowitz. *The Journal of Consumer Affairs*, 36(1), 99–114. <https://doi.org/10.1111/j.1745-6606.2002.tb00422.x>
- Hu, G., & Lei, J. (2015). Chinese University students' perceptions of plagiarism. *Ethics & Behavior*, 25(3), 233–255. <https://doi.org/10.1080/10508422.2014.923313>
- Hu, G., & Lei, J. (2016). Plagiarism in English academic writing: A comparison of Chinese university teachers' and students' understandings and stances. *System*, 56, 107–118. <https://doi.org/10.1016/j.system.2015.12.003>
- Huang, R., Zheng, X., Shang, Y., & Xue, X. (2023). On challenges of AI to cognitive security and safety. *Security and Safety*, 2, 2023012. <https://doi.org/10.1051/sands/2023012>
- Hung, T. C., & Hung, T. W. (2020). How China's cognitive warfare works. *Journal of Global Security Studies*, 7(4). <https://doi.org/10.1093/jogss/ogac016>
- Hunt, S. D. (1993). Objectivity in marketing theory and research. *Journal of Marketing*, 57(2), 76–91. <https://doi.org/10.1177/002224299305700206>
- Hunt, S. D. (2016). *Controversy in marketing theory: For reason, realism, truth and objectivity: For reason, realism, truth and objectivity*. Routledge.
- Hunt, S. D. (2018). Advancing marketing strategy in the marketing discipline and beyond: From promise, to neglect, to prominence, to fragment (to promise?). *Journal of Marketing Management*, 34(1-2), 16–51. <https://doi.org/10.1080/0267257X.2017.1326973>
- Hunt, S. D., & Chonko, L. B. (1984). Marketing and machiavellianism. *Journal of Marketing*, 48(3), 30–42. <https://doi.org/10.2307/1251327>
- Hunt, S. D., & Hansen, J. M. (2007). Understanding ethical diversity in organizations. *Organizational Dynamics*, 36(2), 202–216. <https://doi.org/10.1016/j.orgdyn.2007.03.007>
- Hunt, S. D., Hass, A., & Manis, K. T. (2021). The five stages of the macromarketing field of study: From raison d'être to field of significant promise. *Journal of Macromarketing*, 41(1), 10–24. <https://doi.org/10.1177/0276146721990987>
- Hunt, S. D., & Vitell, S. (1986). A general theory of marketing ethics. *Journal of Macromarketing*, 6(1), 5–16. <https://doi.org/10.1177/027614678600600103>
- Hunt, S. D., & Vitell, S. J. (2006). The general theory of marketing ethics: A revision and three questions. *Journal of Macromarketing*, 26(2), 143–153. <https://doi.org/10.1177/0276146706290923>
- Hunter, L. Y., Albert, C. D., Rutland, J., Topping, K., & Hennigan, C. (2024). Artificial intelligence and information warfare in major power states: How the US, China, and Russia are using artificial intelligence in their information warfare and influence operations. *Defense & Security Analysis*, 40(2), 1–35. <https://doi.org/10.1080/14751798.2024.2321736>
- Hyman, M. R. (2024). Freeing social and medical scientists from the replication crisis. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4898637
- Hyman, M. R., & Kostyk, A. (2019). Guest editorial: A prospectus on marketing futurology. *European Journal of Marketing*, 53(8), 1485–1503. <https://doi.org/10.1108/EJM-08-2019-968>
- Hyman, M. R., & Kostyk, A. (2022). An aspirational definition of marketing: An abstract. In F. Pantoja & S. Wu (Eds.), *From micro to macro: Dealing with uncertainties in the global marketplace. AMSAC 2020. Developments in marketing science: Proceedings of the academy of marketing science*. Springer. https://doi.org/10.1007/978-3-030-89883-0_75
- Hyman, M. R., & Lukosius, V. (2024). Refocusing and futuring perspectives on AI in marketing. *Journal of Macromarketing*, 44(4), 906–919. <https://doi.org/10.1177/02761467241280364>
- Ibrahim, H., Liu, F., Asim, R., Battu, B., Benabderrahmane, S., Alhafni, B., Adnan, W., Alhanai, T., AlShebli, B., Baghdadi, R., Bélanger, J. J., Beretta, E., Celik, K., Chaqfeh, M., Daqaq, M. F., Bernoussi, Z. E., Fougne, D., . . . Zaki, Y. (2023). Perception, performance, and detectability of conversational artificial intelligence across 32 university courses. *Scientific Reports*, 13(1), 12187. <https://doi.org/10.48550/arXiv.2305.13934>
- Jha, A. (2012, September 13). False positives: Fraud and misconduct are threatening scientific research. *The Guardian*. <https://www.theguardian.com/science/2012/sep/13/scientific-research-fraud-bad-practice>
- Jian, H., Marion, R., & Wang, W. (2019). Academic integrity from China to the United States: The acculturation process for Chinese graduate students in the United States. *Ethics & Behavior*, 29(1), 51–70. <https://doi.org/10.1080/10508422.2018.1468760>
- Johnson-Hall, T. D. (2017). Ensuring food safety by preventing food recalls: The impact of locus of failure, regulatory agency discovery, breadth, and firm size on corrective action. *Journal of Marketing Channels*, 24(3-4), 115–135. <https://doi.org/10.1080/1046669X.2017.1393230>
- Jones, T. M. (1991). Ethical decision making by individuals in organizations: An issue-contingent model. *Academy of Management Review*, 16(2), 366–395. <https://doi.org/10.5465/amr.1991.4278958>
- Kadel, R., Mishra, B. K., Shailendra, S., Abid, S., Rani, M., & Mahato, S. P. (2024, July). Crafting tomorrow's evaluations: Assessment design strategies in the era of generative AI. In *2024 International symposium on educational technology (ISET)* (pp. 13–17). IEEE.
- Kaushal, A., Altman, R., & Langlotz, C. (2020). Geographic distribution of US cohorts used to train deep learning algorithms. *JAMA*, 324(12), 1212–1213. <https://doi.org/10.1001/jama.2020.12067>
- Kenwright, B. (2024). Is it the end of undergraduate dissertations?: Exploring the advantages and challenges of generative AI models in education. In S. Hai-Jew (Ed.), *Generative AI in teaching and learning* (pp. 46–65). IGI Global. <https://doi.org/10.4018/979-8-3693-0074-9.ch003>
- Kopalle, P. K., Gangwar, M., Kaplan, A., Ramachandran, D., Reinartz, W., & Rindfleisch, A. (2022). Examining artificial intelligence (AI) technologies in marketing via a global lens: Current trends and future research opportunities. *International Journal of Research in Marketing*, 39(2), 522–540. <https://doi.org/10.1016/j.ijresmar.2021.11.002>

- Kotler, P. (2024, March 12). A lifetime in marketing: Lessons learned and the way ahead, by Philip Kotler. *American Marketing Association*. <https://www.ama.org/2024/03/12/a-lifetime-in-marketing-lessons-learned-and-the-way-ahead-by-philip-kotler/>
- Kotler, P., & Levy, S. J. (1969). Broadening the concept of marketing. *Journal of Marketing*, 33(1), 10–15. <https://doi.org/10.2307/1248740>
- Laczniak, G., & Shultz, C. (2021). Toward a doctrine of socially responsible marketing (SRM): A macro and normative-ethical perspective. *Journal of Macromarketing*, 41(2), 201–231. <https://doi.org/10.1177/0276146720963682>
- Laczniak, G. R., & Klein, T. A. (2010). Caritas in veritate: Updating Catholic social teachings for macromarketing and business strategy. *Journal of Macromarketing*, 30(3), 293–296. <https://doi.org/10.1177/0276146710372224>
- Laczniak, G. R., & Murphy, P. E. (2006). Normative perspectives for ethical and socially responsible marketing. *Journal of Macromarketing*, 26(2), 154–177. <https://doi.org/10.1177/0276146706290924>
- Laczniak, G. R., & Murphy, P. E. (2012). Stakeholder theory and marketing: Moving from a firm-centric to a societal perspective. *Journal of Public Policy & Marketing*, 31(2), 284–292. <https://doi.org/10.1509/jppm.10.106>
- Lakshmi, R. (2015). These Indian parents climbed a school wall to help their kids cheat on an exam. *The Washington Post*. March 19, Accessed at. <https://www.washingtonpost.com/news/worldviews/wp/2015/03/19/these-indian-parents-climbed-a-school-wall-to-help-their-kids-cheat-on-an-exam/>
- Latan, H., Chiappetta Jabbour, C. J., & Lopes de Sousa Jabbour, A. B. (2019). ‘Whistleblowing triangle’: Framework and empirical evidence. *Journal of Business Ethics*, 160, 189–204. <https://doi.org/10.1007/s10551-018-3862-x>
- Latino, M. E., Corallo, A., Menegoli, M., & Nuzzo, B. (2022). An integrative conceptual framework of food certifications: Systematic review, research agenda, and macromarketing implications. *Journal of Macromarketing*, 42(1), 71–99. <https://doi.org/10.1177/02761467211049071>
- Legal Information Institute. (2025). *Daubert Standard*. Cornell Law School. https://www.law.cornell.edu/wex/daubert_standard
- LeMesurier, J. L. (2017). Uptaking race: Genre, MSG, and Chinese dinner. *Poroi*, 12(2), 1–23. <https://doi.org/10.13008/2151-2957.1253>
- Lewis-Kraus, G. (2023, October 9). They studied dishonesty. Was their work a lie? *The New Yorker*. <https://www.newyorker.com/magazine/2023/10/09/they-studied-dishonesty-was-their-work-a-lie>
- Liu, X., & Chen, X. (2018). Journal retractions: Some unique features of research misconduct in China. *Journal of Scholarly Publishing*, 49(3), 305–319. <https://doi.org/10.3138/jsp.49.3.02>
- Longpre, S., Mahari, R., Lee, A., Lund, C., Oderinwale, H., Brannon, W., Saxena, N., Obeng-Marnu, N., South, T., Hunter, C., Klyman, K., Klamm, C., Schoelkopf, H., Singh, N., Cherep, M., Anis, A., Dinh, A., Chitongo, C., . . . Pentland, A. (2024). Consent in crisis: The rapid decline of the AI data commons. <https://arxiv.org/abs/2407.14933>
- Ma, L., & Sun, B. (2020). Machine learning and AI in marketing—connecting computing power to human insights. *International Journal of Research in Marketing*, 37(3), 481–504. <https://doi.org/10.1016/j.ijresmar.2020.04.005>
- MacDonald, A., & Ratcliffe, R. (2023). Cognitive warfare: Maneuvering in the human dimension. U.S. Naval Institute. Proceedings, 149/4, 1. <https://www.usni.org/magazines/proceedings/2023/april/cognitive-warfare-maneuvering-human-dimension>
- Magnus, J. R., Polterovich, V. M., Danilov, D. L., & Savvateev, A. V. (2002). Tolerance of cheating: An analysis across countries. *The Journal of Economic Education*, 33(2), 125–135. <https://doi.org/10.1080/00220480209596462>
- Májovský, M., Černý, M., Kasal, M., Komarc, M., & Netuka, D. (2023). Artificial intelligence can generate fraudulent but authentic-looking scientific medical articles: Pandora’s box has been opened. *Journal of Medical Internet Research*, 25(1), e46924. <https://doi.org/10.2196/46924>
- Maloshonok, N., & Shmeleva, E. (2019). Factors influencing academic dishonesty among undergraduate students at Russian universities. *Journal of Academic Ethics*, 17(3), 313–329. <https://doi.org/10.1007/s10805-019-9324-y>
- Manning, L. (2017). The influence of organizational subcultures on food safety management. *Journal of Marketing Channels*, 24(3–4), 180–189. <https://doi.org/10.1080/1046669X.2017.1393235>
- Mariani, M. M., Perez-Vega, R., & Wirtz, J. (2022). AI In marketing, consumer research and psychology: A systematic literature review and research agenda. *Psychology & Marketing*, 39(4), 755–776. <https://doi.org/10.1002/mar.21619>
- Marken, S. (2019, April 12). A crisis of confidence in higher ed. *Gallup*. <https://news.gallup.com/opinion/gallup/248492/crisis-confidence-higher.aspx>
- McQuarrie, E. F. (2014). Threats to the scientific status of experimental consumer psychology: A Darwinian perspective. *Marketing Theory*, 14(4), 477–494. <https://doi.org/10.1177/1470593114540678>
- Mehta, P., Jebarajakirthy, C., Maseeh, H. I., Anubha, A., Saha, R., & Dhanda, K. (2022). Artificial intelligence in marketing: A meta-analytic review. *Psychology & Marketing*, 39(11), 2013–2038. <https://doi.org/10.1002/mar.21716>
- Moody, J. (2024, January 12). When presidents plagiarize. *Insider Higher Ed*. <https://www.insidehighered.com/news/governance/executive-leadership/2024/01/12/when-college-presidents-plagiarize>
- Moorman, C., & Day, G. S. (2016). Organizing for marketing excellence. *Journal of Marketing*, 80(6), 6–35. <https://doi.org/10.1509/jm.15.0423>
- Moorman, C., Deshpande, R., & Zaltman, G. (1993). Factors affecting trust in market research relationships. *Journal of Marketing*, 57(1), 81–101. <https://doi.org/10.1177/002224299305700106>
- Morgan, R. M., & Hunt, S. D. (1994). The commitment-trust theory of relationship marketing. *Journal of Marketing*, 58(3), 20–38. <https://doi.org/10.1177/002224299405800302>
- Motta, M., & Stecula, D. (2021). Quantifying the effect of Wakefield et al. (1998) on skepticism about MMR vaccine safety in the U.S. *PLoS ONE*, 16(8). <https://doi.org/10.1371/journal.pone.0256395>
- Moussa, S. (2021). A ‘Trojan Horse’ in the reference lists: Citations to a hijacked journal in SSCI-indexed marketing journals. *The Journal of Academic Librarianship*, 47(5), 102388. <https://doi.org/10.1016/j.jalib.2021.102388>

- MSI. (2024). Research priorities 2024. *Marketing Science Institute*. <https://www.msi.org/wp-content/uploads/2024/04/2024-RP.pdf>
- Mu, J., & Zhang, J. Z. (2025). Artificial intelligence marketing usage and firm performance. *Journal of the Academy of Marketing Science*, 1–54. <https://doi.org/10.1007/s11747-024-01076-z>
- Murphy, P. E. (2017). Marketing ethics at the millennium: Review, reflections, and recommendations. *The Blackwell Guide to Business Ethics*, 165–185. <https://doi.org/10.1002/9781405164771.ch8>
- Murphy, P. E., Lacznia, G. R., & Wood, G. (2007). An ethical basis for relationship marketing: A virtue ethics perspective. *European Journal of Marketing*, 41(1/2), 37–57. <https://doi.org/10.1108/03090560710718102>
- NCES. (2020). Fast facts: Educational institutions. *National Center for Education Statistics*. <https://nces.ed.gov/fastfacts/display.asp?id=1122>
- Newman, C. L., Turri, A. M., Howlett, E., & Stokes, A. (2014). Twenty years of country-of-origin food labeling research: A review of the literature and implications for food marketing systems. *Journal of Macromarketing*, 34(4), 505–519. <https://doi.org/10.1177/0276146714529306>
- Nietzel, M. T. (2024, July 9). Americans' confidence in higher education drops again, finds Gallup. *Forbes*. <https://www.forbes.com/sites/michaelt Nietzel/2024/07/09/americans-confidence-in-higher-education-drops-again-finds-gallup/>
- Nill, A., & Schibrowsky, J. A. (2007). Research on marketing ethics: A systematic review of the literature. *Journal of Macromarketing*, 27(3), 256–273. <https://doi.org/10.1177/0276146707304733>
- Nills, A. L., & Shultz, C. J. (1997). Marketing ethics across cultures: Decision-making guidelines and the emergence of dialogic idealism. *Journal of Macromarketing*, 17(2), 4–19. <https://doi.org/10.1177/027614679701700202>
- O'Connor, A. (2018, September 29). More evidence that nutrition studies don't always add up. *New York Times*. <https://www.nytimes.com/2018/09/29/sunday-review/cornell-food-scientist-wansink-misconduct.html>
- Ooi, K. B., Tan, G. W. H., Al-Emran, M., Al-Sharafi, M. A., Capatina, A., Chakraborty, A., Dwivedi, Y. K., Huang, T.-L., Kar, A. K., Lee, V.-H., Loh, X.-M., Micu, A., Mikalef, P., Mogaji, E., Pandey, N., Raman, R., & Rana, N. P., . . . L. W. Wong (2025). The potential of generative artificial intelligence across disciplines: Perspectives and future directions. *Journal of Computer Information Systems*, 65(1), 76–107. <https://doi.org/10.1080/08874417.2023.2261010>
- Oransky, I., & Redman, B. (2024). Rooting out scientific misconduct. *Science (New York, N Y)*, 383(6679), 131. <https://doi.org/10.1126/science.adn9352>
- Østergaard, S. D., & Nielbo, K. L. (2023). False responses from artificial intelligence models are not hallucinations. *Schizophrenia Bulletin*, 49(5), 1105–1107. <https://doi.org/10.1093/schbul/sbad068>
- Owens, K. (2024, February 27). How long does it take to write a research paper? *Medium*. <https://medium.com/@kieran-listening/how-long-does-it-take-to-write-a-research-paper-317fcb6cf6d2>
- Pan, J., & Lei, J. (2024). Profiling Chinese university students' understandings of plagiarism through Q methodology. *Ethics & Behavior*, 1–16. <https://doi.org/10.1080/10508422.2024.2389541>
- Park, D., Schivinski, B., Duns-McKay, H., Heeren, B., & McLachlan, E. (2025). The effect of social media communication on intention to vaccinate against COVID-19: A relationship mediated by perceptions of vaccine efficacy and safety. *Journal of Marketing Communications*, 31(1), 1–20. <https://doi.org/10.1080/13527266.2023.2191618>
- Paschen, J., Kietzmann, J., & Kietzmann, T. C. (2019). Artificial intelligence (AI) and its implications for market knowledge in B2B marketing. *Journal of Business & Industrial Marketing*, 34(7), 1410–1419. <https://doi.org/10.1108/JBIM-10-2018-0295>
- Payan, J., Reardon, J., & McCorkle, D. E. (2010). The effect of culture on the academic honesty of marketing and business students. *Journal of Marketing Education*, 32(3), 275–291. <https://doi.org/10.1177/0273475310377781>
- Paygah, M. R., & Colabi, A. M. (2023). Thematic analysis of direct-to-consumer advertising (DTCA) for vaccination campaigns during the COVID-19 pandemic: Healthcare experts' insights. *Journal of Marketing Communications*, 1–32. <https://doi.org/10.1080/13527266.2023.2289615>
- Peres, R., Schreier, M., Schweidel, D., & Sorescu, A. (2023). On ChatGPT and beyond: How generative artificial intelligence may affect research, teaching, and practice. *International Journal of Research in Marketing*, 40(2), 269–275. <https://doi.org/10.1016/j.ijresmar.2023.03.001>
- Peterson, M. (2006). Focusing the future of macromarketing. *Journal of Macromarketing*, 26(2), 245–249. <https://doi.org/10.1177/0276146706291047>
- Piper, K. (2024, March 1). Who fakes cancer research? Apparently, lots of people. *Vox*. <https://www.vox.com/future-perfect/24086809/fake-cancer-research-data-scientific-fraud>
- Polonsky, M. J., & Rotman, J. D. (2023). Should artificial intelligent agents be your co-author? Arguments in favour, informed by ChatGPT. *Australasian Marketing Journal*, 31(2), 91–96. <https://doi.org/10.1177/14413582231167882>
- Quan, W., Chen, B., & Shu, F. (2017). Publish or impoverish: An investigation of the monetary reward system of science in China (1999–2016). *Aslib Journal of Information Management*, 69(5), 486–502. <https://doi.org/10.1108/AJIM-01-2017-0014>
- Rane, N. L., Choudhary, S. P., Tawde, A., & Rane, J. (2023). ChatGPT is not capable of serving as an author: Ethical concerns and challenges of large language models in education. *International Research Journal of Modernization in Engineering Technology and Science*, 5(10), 851–874. <https://doi.org/10.56726/IRJMETS45212>
- Ransbotham, S., Fichman, R. G., Gopal, R., & Gupta, A. (2016). Special section introduction—Ubiquitous IT and digital vulnerabilities. *Information Systems Research*, 27(4), 834–847. <https://doi.org/10.1287/isre.2016.0683>
- Ravi, A., & Paulus, V. H. (2024, April 17). Harvard med school professor plagiarized report. https://www.realcleareducation.com/2024/04/17/harvard_med_school_professor_plagiarized_report_1025678.html
- Rawwas, M. Y., Al-Khatib, J. A., & Vitell, S. J. (2004). Academic dishonesty: A cross-cultural comparison of US and Chinese marketing students. *Journal of Marketing Education*, 26(1), 89–100. <https://doi.org/10.1177/0273475303262354>

- Rawwas, M. Y., Swaidan, Z., & Al-Khatib, J. (2006). Does religion matter? A comparison study of the ethical beliefs of marketing students of religious and secular universities in Japan. *Journal of Business Ethics*, 65, 69–86. <https://doi.org/10.1007/s10551-006-0023-4>
- Reibstein, D. J., Day, G., & Wind, J. (2009). Guest editorial: Is marketing academia losing its way? *Journal of Marketing*, 73(4), 1–3. <https://www.jstor.org/stable/20619030> <https://doi.org/10.1509/jmkg.73.4.001>
- Rindfleisch, A., Kim, M., & Lee, E. J. H. (2024). AI as a convivial tool. *Journal of Macromarketing*, 44(4), 920–927. <https://doi.org/10.1177/02761467241285322>
- Rodgers, W., Söderbom, A., & Guiral, A. (2015). Corporate social responsibility enhanced control systems reducing the likelihood of fraud. *Journal of Business Ethics*, 131, 871–882. <https://doi.org/10.1007/s10551-014-2152-5>
- Roose, K. (2024, July 19). The data that powers A.I. is disappearing fast. <https://www.nytimes.com/2024/07/19/technology/ai-data-restrictions.html>
- Ross, C. G. (2015). Toward a new consensus for tenure in the twenty-first century. *AAUP*, May–June. <https://www.aaup.org/article/toward-new-consensus-tenure-twenty-first-century>
- Saba, W. S. (2024). LLMs' understanding of natural language revealed. *arXiv preprint arXiv:2407.19630*. <https://doi.org/10.48550/arXiv.2407.19630>
- Sabel, B. A., Knaack, E., Gigerenzer, G., & Bilc, M. (2023). Fake publications in biomedical science: Red-flagging method Indicates mass production. *MedRxiv*, 2023-05. <https://doi.org/10.1101/2023.05.06.23289563>
- Scheinbaum, A. C., Steinfeld, L., Giebelhausen, M., & Dobscha, S. (2024). A transformative orientation model for encouraging responsible marketing actions. *Journal of Macromarketing*, 02761467241269832, <https://doi.org/10.1177/02761467241269832>
- Schlegelmilch, B. B., & Öberseder, M. (2010). Half a century of marketing ethics: Shifting perspectives and emerging trends. *Journal of Business Ethics*, 93, 1–19. <https://doi.org/10.1007/s10551-009-0182-1>
- Schneider, B., Alexander, J., & Thomas, P. (2023). Publications output: U.S. trends and international comparisons. *National Science Foundation*. <https://ncses.nsf.gov/pubs/nsb202333/publication-output-by-region-country-or-economy-and-by-scientific-field#output-by-region-country-or-economy>
- Schwarzschild, A., Goldblum, M., Gupta, A., Dickerson, J. P., & Goldstein, T. (2021, July). Just how toxic is data poisoning? A unified benchmark for backdoor and data poisoning attacks. International conference on machine learning. (pp. 9389–9398). PMLR. <https://proceedings.mlr.press/v139/schwarzschild21a.html>
- Seo, Y., Kim, H., Ko, Y., Yoon, H. J., Han, J. Y., Lee, J., & Seo, J. K. (2023). The power of “likes”: The effects of “liked by” and number of likes cues on perceived descriptive norms and behavioral intention to receive the flu vaccine. *Journal of Marketing Communications*, 1–17. <https://doi.org/10.1080/13527266.2023.2235753>
- Shabbir, H. A., Hyman, M. R., & Kostyk, A. (2021). A macromarketing prescription for COVID-19: Solidarity and care ethics. *Journal of Macromarketing*, 41(2), 181–193. <https://doi.org/10.1177/02761467211001544>
- Shankar, V. (2024). Managing the twin faces of AI: A commentary on “is AI changing the world for better or worse?”. *Journal of Macromarketing*, 44(4), 892–899. <https://doi.org/10.1177/02761467241286483>
- Sharma, S. (2024). Benefits or concerns of AI: A multistakeholder responsibility. *Futures*, 157, 103328. <https://doi.org/10.1016/j.futures.2024.103328>
- Sheth, J. N., & Parvatiyar, A. (2022). Socially responsible marketing: Toward aligning dharma (duties), karma (actions), and eudaimonia (well-being). *Journal of Macromarketing*, 42(4), 590–602. <https://doi.org/10.1177/02761467221130260>
- Singhapakdi, A., & Vitell, S. J. (1990). Marketing ethics: Factors influencing perceptions of ethical problems and alternatives. *Journal of Macromarketing*, 10(1), 4–18. <https://doi.org/10.1177/027614679001000102>
- Sirgy, M. J. (2025). How can marketing address the big issues of our time. *Journal of Macromarketing*, 45(1), 02761467241310857. <https://doi.org/10.1177/02761467241310857>
- Stamatiadis-Bréhier, A. (2024). The power of second-order conspiracies. *Inquiry*, 1–26. <https://doi.org/10.1080/0020174X.2024.2375781>
- Stern, J. (2023, August 2). An unsettling hint at how much fraud could exist in science. *The Atlantic*. <https://www.theatlantic.com/science/archive/2023/08/gino-ariely-data-fraud-allegations/674891/>
- Stević, A. (2024, July 17). Scopus is broken – just look at its literature category. *Retractionwatch.com* <https://retractionwatch.com/2024/07/17/scopus-is-broken-just-look-at-its-literature-category/#more-129655>
- Studnicki, J., Harrison, D. J., Longbons, T., Skop, I., Reardon, D. C., Fisher, J. W., Tsulukidze, M., & Craver, C. (2021). RETRACTED: A longitudinal cohort study of emergency room utilization following mifepristone chemical and surgical abortions, 1999–2015. *Health Services Research and Managerial Epidemiology*, 8, <https://doi.org/10.1177/23333928211053965>
- Studnicki, J., Longbons, T., Fisher, J. W., Harrison, D. J., Skop, I., & MacKinnon, S. J. (2019). RETRACTED: Doctors who perform abortions: Their characteristics and patterns of holding and using hospital privileges. *Health Services Research and Managerial Epidemiology*, 6, <https://doi.org/10.1177/2333392819841211>
- Studnicki, J., Longbons, T., Harrison, D. J., Skop, I., Cirucci, C., Reardon, D. C., Craver, C., Fisher, J. W., & Tsulukidze, M. (2022). RETRACTED: A post hoc exploratory analysis: Induced abortion complications mistaken for miscarriage in the emergency room are a risk factor for hospitalization. *Health Services Research and Managerial Epidemiology*, 9, <https://doi.org/10.1177/23333928221103107>
- Thiese, M. S., Walker, S., & Lindsey, J. (2017). Truths, lies, and statistics. *Journal of Thoracic Disease*, 9(10), 4117–4124. <https://doi.org/10.21037/jtd.2017.09.24>
- Thormundsson, B. (2023). Usage of generative AI in the U.S. 2023. <https://www.statista.com/statistics/1413836/use-of-generative-ai-us/>
- Thyroff, A., Hawkins, M. A., & Akdevelioglu, D. (2023). Thinking big about going small: Conceptualizing the human-technology integration spectrum. *Journal of Macromarketing*, 43(4), 433–440. <https://doi.org/10.1177/02761467231191078>

- Tian, M., & Lu, G. (2017). What price the building of world-class universities? Academic pressure faced by young lecturers at a research-centered university in China. *Teaching in Higher Education*, 22(8), 957–974. <https://doi.org/10.1080/13562517.2017.1319814>
- Times of India. (2024, August 28). Indian-origin tech influencer Debarghya Das under fire for this suggestion to get US citizenship. *Times of India*. <https://timesofindia.indiatimes.com/world/us/indian-origin-tech-influencer-under-fire-for-this-suggestion-to-get-us-citizenship/articleshow/112844695.cms>
- Truong, V. D. (2017). Government-led macro-social marketing programs in Vietnam: Outcomes, challenges, and implications. *Journal of Macromarketing*, 37(4), 409–425. <https://doi.org/10.1177/0276146716660833>
- Underwood, K. (2017, August 15). Cognitive warfare will be deciding factor in battle. *Signal*. <https://www.afcea.org/signal-media/cyber/cognitive-warfare-will-be-deciding-factor-battle>
- Urseu, T. (2023). Is generative artificial intelligence a black swan in cognitive warfare? In *Proceedings of the intentional scientific conference strategies* (Vol. XIX, pp. 284–288).
- van Dalen, H. P. (2021). How the publish-or-perish principle divides a science: The case of economists. *Scientometrics*, 126(2), 1675–1694. <https://doi.org/10.1007/s11192-020-03786-x>
- Van Noorden, R. (2023). More than 10,000 research papers were retracted in 2023—a new record. *Nature*, 624(7992), 479–481. <https://doi.org/10.1038/d41586-023-03974-8>
- Venkatesh, A., & Peñaloza, L. (2015). From marketing to the market: A call for paradigm shift. In J. N. Sheth & R. S. Sisodia (Eds.), *Does marketing need reform? Fresh perspectives on the future* (pp. 142–158). Routledge.
- Wakefield, A. J., Murch, S. H., Anthony, A., Linnell, J., Casson, D. M., Malik, M., Berelowitz, M., Dhillon, A. P., Thomson, M. A., Harvey, P., Valentine, A., Davies, S. E., & Walker-Smith, J. A. (1998). RETRACTED: Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children. *The Lancet*, 351(9103), 637–641. [https://doi.org/10.1016/S0140-6736\(97\)11096-0](https://doi.org/10.1016/S0140-6736(97)11096-0)
- Wang, L., Touré-Tillery, M., & McGill, A. L. (2023). The effect of disease anthropomorphism on compliance with health recommendations. *Journal of the Academy of Marketing Science*, 51(2), 266–285. <https://doi.org/10.1007/s11747-022-00891-6>
- Wang, T., Li, L., Chen, X., & Li, K. (2024). A study on the risks and countermeasures of false information caused by AIGC. *Journal of Electrical Systems*, 20(3), 420–426. <https://doi.org/10.52783/jes.2866>
- Watson, F., & Wu, Y. (2022). The impact of online reviews on the information flows and outcomes of marketing systems. *Journal of Macromarketing*, 42(1), 146–164. <https://doi.org/10.1177/02761467211042552>
- Weber, T. J., Muehling, D. D., & Kareklas, I. (2021). How unsponsored, online user-generated content impacts consumer attitudes and intentions toward vaccinations. *Journal of Marketing Communications*, 27(4), 389–414. <https://doi.org/10.1080/13527266.2019.1671479>
- Wilkinson, L. (2023, August 8). JLL rolls out proprietary generative AI model to internal employees. <https://www.ciodive.com/news/JLL-GPT-Technologies-rollout-generative-AI/690146/>
- Woolley, S. (2020, January 8). We're fighting fake news AI bots by using more AI. That's a mistake. <https://www.technologyreview.com/2020/01/08/130983/were-fighting-fake-news-ai-bots-by-using-more-ai-thats-a-mistake/>
- Xie, Y., Zhang, C., & Lai, Q. (2014). China's rise as a major contributor to science and technology. *Proceedings of the National Academy of Sciences of the United States of America*, 111(26), 9437–9442. <https://doi.org/10.1073/pnas.1407709111>
- Xu, J., Duan, X., & Ding, Y. (2018). The changing geography of business and management research in Asia. *Scientometrics*, 117(1), 65–87. <https://doi.org/10.1177/0308518X18816694>
- Yang, W. (2013). Research integrity in China. *Science (New York, N Y)*, 342(6162), 1019. <https://doi.org/10.1126/science.1247700>
- Yong, E. (2012, July 12). Uncertainty shrouds psychologist's resignation. *Nature*. <https://doi.org/10.1038/nature.2012.10968>
- Zhaksylyk, A., Zimba, O., Yessirkepov, M., & Kocyigit, B. F. (2023). Research integrity: Where we are and where we are heading. *Journal of Korean Medical Science*, 38(47), 1–11. <https://doi.org/10.3346/jkms.2023.38.e405>
- Zhang, L. (2023). Why is suicide incidence among doctoral students so high?—A study of China's doctoral tutorial system. *Heliyon*, 9(8). <https://doi.org/10.1016/j.heliyon.2023.e18382>
- Zimmermann, L., Somasundaram, J., & Saha, B. (2024). Adoption of new technology vaccines. *Journal of Marketing*, 88(4), 1–21. <https://doi.org/10.1177/00222429231220295>

Copyright of Journal of Macromarketing is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.